A
Project Report on

UNIFIED AI AGENT FOR AUTOMATING TASKS ACROSS GOOGLE WORKSPACE

Submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING (DATA SCIENCE)

By

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Under the esteemed guidance of

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (DATA SCIENCE)
Geethanjali College of Engineering and Technology
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Accredited by NAAC with A+ Grade; B.Tech. CSE, EEE, ECE accredited by NBA
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CERTIFICATE

This is to certify that the project report entitled **“UNIFIED AI AGENT FOR AUTOMATING TASKS ACROSS GOOGLE WORKSPACE”** is a bonafide work done by Thimmapuram Srinivas Chary **(22R11A6739),** Dyavari Ramakrishna **(23R15A6702),** Kokkalakonda Akhil **(22R11A6723)**, in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in “**Computer Science and Engineering (Data Science)**” from Jawaharlal Nehru Technological University, Hyderabad during the year 2025-2026.

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DECLARATION BY THE CANDIDATES

We, **Thimmapuram Srinivas Chary,** bearing Roll No. **22R11A6739, Dyavari Ramakrishna,** bearing Roll No. **23R15A6702,** and **Kokkalakonda Akhil** bearing Roll No. **22R11A6723**, hereby declare that the project report entitled **"Unified AI agent for automating tasks across google workspace"** is done under the guidance of **Mr. S. Tirupathi Rao, Associate Professor**, Department of Computer Science and Engineering (Data Science), Geethanjali College of Engineering and Technology, is submitted in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in CSE(Data Science)**.

This is a record of bonafide work carried out by us in **Geethanjali College of Engineering and Technology** and the results embodied in this project have not been reproduced or copied from any source. The results embodied in this project report have not been submitted to any other University or Institute for the award of any other degree or diploma.

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Correlation Between The Major Project and the Program Outcomes
(Po's, Program Specific Outcomes (Pso's)

IV YEAR II SEM SECTION – A **BATCH NO: 7**		**TITLE OF THE PROJECT**
ROLL NO	**NAME**	Unified AI agent for automating tasks across google workspace
22R11A6739	Thimmapuram Srinivas Chary	
23R15A6702	Dyavari Ramakrishna	
22R11A6723	Kokkalakonda Akhil	

Table 1: Major project correlation with appropriate PO's and PSO's
(Please specify level of correlation, High – 3, Medium – 2 and Low – 1)

PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PSO 1	PSO 2	PSO 3

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Sustainable Development Goals (SDG) Table for Unified AI agent for automating tasks across google workspace

SDG No.	SDG NAME	Relevance to your project	How your System contributes
SDG 4	Quality Education	Supports easy learning through automation and NLP.	Helps users manage tasks, documents, and schedules efficiently.
SDG 8	Decent Work and Economic Growth	Improves productivity by reducing manual work.	Automates emails, scheduling, and reporting tasks.
SDG 9	Industry, Innovation and Infrastructure	Uses AI and integration technologies.	Builds a smart system connecting multiple applications.
SDG 10	Reduced Inequalities	Simple interface accessible to all users.	Enables non-technical users to use advanced tools easily.
SDG 11	Sustainable Cities and Communities	Promotes efficient digital workflows.	Improves task management and reduces complexity.
SDG 12	Responsible Consumption and Production	Reduces unnecessary digital effort.	Minimizes repetitive work and saves resources.
SDG 13	Climate Action	Encourages digital over physical processes.	Supports paperless work using Docs, Drive, etc.

ABSTRACT

This project presents the design and implementation of a Unified AI Agent aimed at automating tasks within the Google Workspace environment, with the primary goal of enhancing productivity and reducing manual effort in everyday digital workflows. The system is developed using Python as the core programming language due to its simplicity, scalability, and extensive ecosystem of libraries that support artificial intelligence, automation, and API integration. Natural Language Processing (NLP) techniques are incorporated using tools such as NLTK and spaCy to enable the system to understand and process user inputs provided in natural language, whether in text or voice form. These inputs are analyzed to identify user intent and extract relevant entities, which are then passed to advanced AI models, including Large Language Models (LLMs), for reasoning and task planning. The AI agent is capable of breaking down complex user requests into smaller, manageable steps and generating structured workflows for execution, allowing it to handle multi-step and real-world scenarios efficiently. The system integrates seamlessly with various Google Workspace services such as Gmail, Google Calendar, Google Drive, and Google Docs through secure APIs, with OAuth 2.0 ensuring authentication and data protection. This integration enables the agent to perform tasks such as sending emails, scheduling meetings, creating and managing documents, and organizing files in real time. A user-friendly interface is provided to facilitate smooth interaction between the user and the system, offering features such as command input, confirmations, notifications, and generated outputs. Additionally, a feedback and monitoring mechanism is included to track task execution, provide real-time updates, and ensure transparency and reliability in system operations. The overall architecture combines hardware, software, cloud services, and intelligent models to create a robust and scalable automation solution. By leveraging AI-driven decision-making and seamless integration with widely used productivity tools, the proposed system significantly improves efficiency, minimizes human intervention, and provides a smart, adaptable solution for modern digital task management.

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LIST OF SYMBOLS AND ABBREVIATIONS

S. No	Symbol/Abbreviation	Full Form
1	AI	Artificial Intelligence
2	NLP	Natural Language Processing
3	LLM	Large Language Model
4	API	Application Programming Interface
5	UI	User Interface
6	UX	User Experience
7	OAuth	Open Authorization
8	IDE	Integrated Development Environment
9	JSON	JavaScript Object Notation
10	HTTP	Hyper Text Transfer Protocol
11	CRUD	Create, Read, Update, Delete

1. INTRODUCTION

1.1 Problem Statement

A large number of organizations now extensively use digital productivity tools like Google Workspace applications, including Gmail, Google Calendar, Google Drive, Google Docs, and Google Sheets, to handle communication, collaboration, and even day-to-day activities. These tools are important in the contemporary workplaces because they allow efficient sharing of information and management of tasks. Nevertheless, even with the existence of such sophisticated platforms, employees still devote a considerable portion of time to doing repetitive and time-consuming jobs like sorting emails, arranging schedule, meeting deadlines, document summarizing, and report preparation. These activities decrease the overall productivity and do not allow users to concentrate on more strategic and high-value work. The absence of a smart automation of various applications is one of the biggest issues with the current systems. The existing automation solutions are tailored to operate within one application and lack smooth integration among various platforms. To illustrate, email automation software can help sort or reply to email messages but cannot integrate well with calendar management or document management. On the same note, scheduling applications can manage the meeting arrangements, but cannot get access to the relevant documents or create summaries. This restriction compels users to toggle between various applications including Gmail, Drive, Calendar, Docs and Sheets to perform one task, which is inefficient and overloads the mind.

Lack of contextual knowledge and smart decision making within conventional systems is another major problem. The majority of automation systems are based on preset rules and do not comprehend the intent of users written in natural language. Thus, they will not be able to process complicated user requests and multi-step processes. An example of this is the request such as, prepare a report out of my most recent spreadsheet and send it to the team and a communication has to be carried out between several services which the current systems do not accomplish effectively.

As such, there is a high demand of a smart, integrated and automated system with the ability to:

List of comprehend user commands written in natural language.

• Divide large tasks into smaller tasks.

• Coordinate and integrate activities in multiple applications in Google Workspace.

• Automate repetitive processes like email, schedule, document summary, and report creation.

• Increase productivity through less manual work and switching of tasks.

• Provide safe and trustworthy access to user data.

This project suggests a Unified AI Agent to Automate the Tasks in Google Workspace to overcome these challenges. The presented system uses AI, NLP and LLM-based reasoning to comprehend user intent and perform tasks on various applications in a consistent way. The system automates workflow by combining services like Gmail, Calendar, Drive, Docs and Sheets into one smart system.

1.2 Objectives

The primary goal of this project is to develop and design a Unified AI Agent to Automating Tasks in Google Workspace that can ease the everyday digital workflow by intelligently managing tasks in several apps. The system will also minimize human effort, enhance productivity, and offer a smarter experience of engaging with tools, such as Gmail, Google Calendar, Drive, Docs, and Sheets, by using Artificial Intelligence.

In an attempt to realize this, the following are specific objectives of the project:

1. To Develop a Cohesive Automation System in a variety of applications.

The former is to develop a unified smart system that can be utilized in various Google Workspace applications rather than considering each application as a standalone entity.

2. To facilitate the Natural Language Interaction with NLP.

The system will be able to comprehend commands that are given to it using simple human language. When using Natural language processing (NLP), the agent is able to understand what the user requires, extract the important information, dates, names or documents and transform them into relevant actions.

3. To Design Smart task planning and execution.

The other goal is to enable the system to think step-by-step. The AI agent ought to be able to divide complex requests into smaller actions and perform them in the right sequence after intelligent reasoning as opposed to having to perform simple tasks only.

4. To Automate Tedious and Repeat Work.

The project aims at cutting down the burden of daily repetitive work like email handling, meeting organizations, document summarization, file arrangements and report generation. The automated nature of these tasks will assist a user save time and concentrate on other more crucial tasks.

5. To Facilitate Cross-Application Workflow.

The system is supposed to deal with tasks that require simultaneous multi-application tasks. As an example, it must be capable of gathering information in one application, processing it in a different application and presenting the outcome in a third application automatically, without the user intervention.

6. To Provide Integration that is safe and dependable.

One of the relevant elements of this project is security. The system will be configured to tightly integrate with Google services through the appropriate authentication mechanisms that will keep the user data secure yet enable the authorized operations to be conducted smoothly.

7. To offer Context-Sensitive and Customized Support.

The AI agent is expected to not only perform commands but also comprehend. The system can be improved with time as it will be able to give more personalized and relevant assistance by learning the behavior of the users and past interactions.

8. To enhance efficiency and save time.

Reducing the amount of time users can spend on switching applications and manually perform some operations is one of the main objectives. The system aids in enhancing efficiency and general productivity in day to day work through automation of tasks.

9. To create a flexible and scalable system.

The system is designed in a manner that it can be expanded in the future. Adding new features, services or integrations is not a complex process as it does not impact the current structure, which makes it applicable to the real-world usage.

10. To give immediate results and customer feedback.

The system must be fast and give the user instant feedback. It also enables users to revise and approve actions prior to final execution, to make sure accuracy and control.

1.3 Scope

The scope of this project describes what the system can do, where it can be applied, and how it assists in real-world applications. It is suggested that the Unified AI Agent to Automate Things in Google Workspace should ease the daily digital life by uniting numerous applications into a single intelligent system. Users do not need to deal with tasks manually in various tools; they can easily deal with all tasks on one smart platform. The primary concept of this project is to enhance productivity and minimize unnecessary work with the help of Artificial Intelligence. The system enables users to work quickly and more effectively by integrating various applications and automation of common tasks. It also offers a smarter means of engaging with tools, and the overall experience is made simpler and easier to use.

1.3.1 Scope with regards to the application coverage

The system has been built to integrate with various Google Workspace applications like Gmail, Google Calendar, Google Drive, Google Docs, and Google Sheets. They are commonly used to communicate, schedule, store data, and manage documents, yet tend to run independently. Due to this, there is a tendency to alternate between applications which can slow down the working process and complicate it. The project unites all these applications in a single system to address this issue.

This will enable users to manage and browse various services in one location without necessarily having to switch between tabs or tools. Consequently, the workflow will be more streamlined and structured. Moreover, the integration of these applications enhances the coordination of the applications. As an example, data in one application can be directly utilized in another without the need to copy and paste data. This saves time, minimizes errors and makes the system more efficient in the case of professional and academic use.

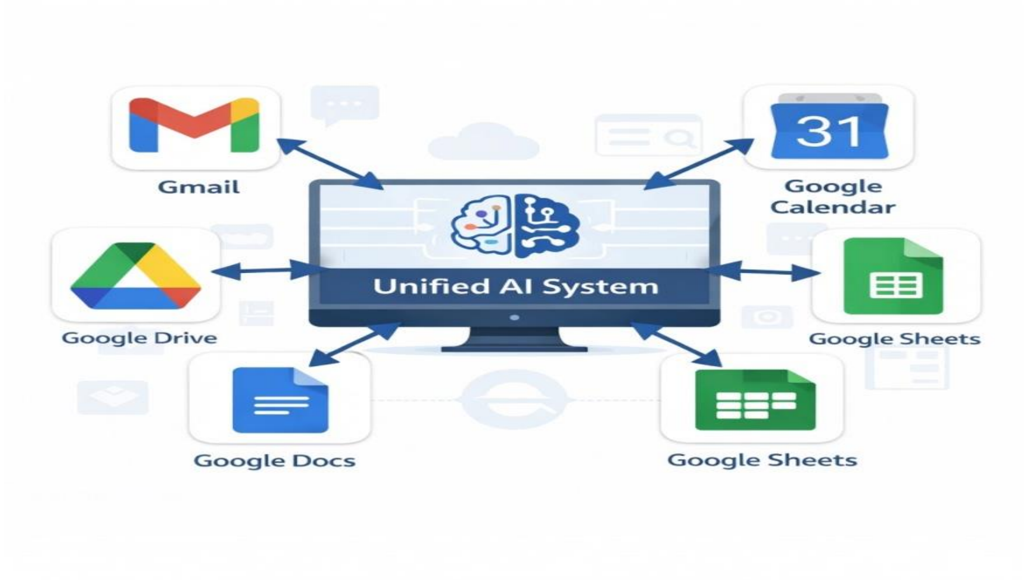

Fig 1.3.1.1 Google Workspace Integration

The diagram above shows how a variety of Google Workspace apps can be integrated into one Unified AI System. It demonstrates the relationship between various services like Gmail, Google Calendar, Google Drive, Google Docs, and Google Sheets and some central intelligent system. All the applications are connected by arrows, which depict how data and communications circulated between the applications and the AI system. The diagram shows how the Unified AI System is a key node that integrates work across various applications. Rather than being able to work alone, each application is connected to the AI system and allows the execution of tasks easily.

1.3.2 Scope of Automation of Tasks.

Automation of routine tasks that a user performs on a daily basis is one of the key objectives of this project. Activities such as handling emails, meeting schedules, file organization, and document summaries usually consume a lot of time in manual handling. The system is geared towards automation of these tasks with just user instructions. The system also saves the workload of users by automating these activities and hence enabling them to concentrate on more significant activities. Users need not have to repeat a certain action over and over again; a system can do it more than once and do it correctly. This enhances productivity and time-saving.

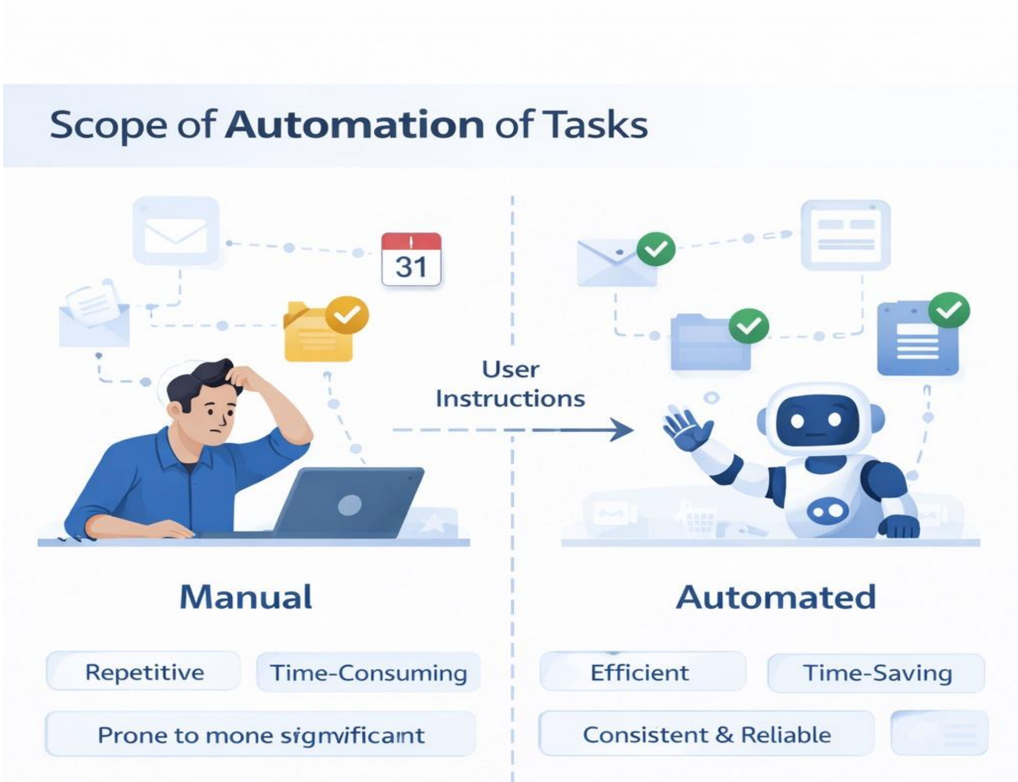

Fig1.3.2.1 Task Automation Overview

The figure above explains the idea of automation of tasks by comparing the manual work with the work via the use of an intelligent system that is automated. The left side of the diagram represents a user operating manually, e.g. working with emails, in schedules and files organization. This side symbolizes the conventional workflows which tend to be repetitive, time consuming and are vulnerable to human errors. On the right, the figure depicts an automated system that does the same things effectively. The system performs various tasks including email management, scheduling and document management with minimum human intervention. The flow representation and arrows show the way the user instructions are handled by the system to automate tasks without a hitch.

1.3.3 Scope in Terms of Natural Language Interaction

The system is constructed in such a way that it is able to comprehend and react to the instructions given in natural language. It implies that users can easily communicate with it in a conversational and easy manner without having to learn complex commands and technical jargons. This renders the system user-friendly to all, even individuals who do not know how to use the advanced technology. Such interaction approach makes the system more intuitive and easy to use. The user is not required to go through various choices but merely provide what they require and the system does all the work. This saves time and simplifies the work with digital tools, making it more comfortable and productive.

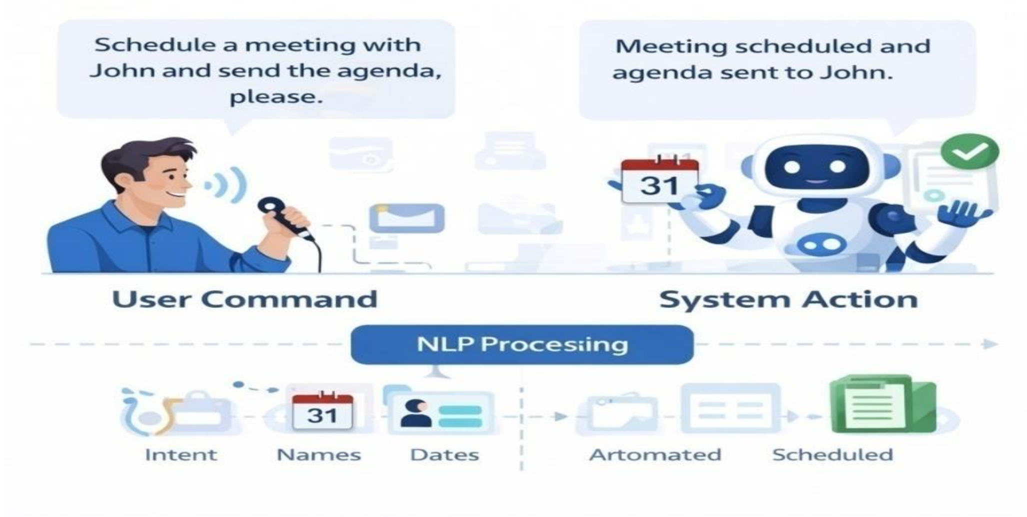

Fig1.3.3.1 Natural Language Interaction

The figure above shows how the system is used by the user in a natural language in a simple and conversational manner. The user issues a command on the left, like scheduling a meeting and sending an agenda, and it is evident that no technical knowledge or complex instructions are needed. This contributes to the ease of use of the system by any form of users. The main section of the diagram is the Natural Language Processing (NLP) component, which involves the system interpreting the input to find significant elements such as intent, names, and dates.

1.3.4 Scope in Terms of Cross-Application Workflow

The system is meant to accommodate activities that entail a number of applications collaborating. In practical scenarios, tasks do not often have only one tool available, and often involve the coordination of other applications. This project aims at simplifying and streamlining such processes. As an example, a user may be required to gather information in Google Sheets, prepare a report in Google Docs and mail it through Gmail. The system is capable of carrying out all these steps automatically one right after another without having to input anything manually at every step. This makes complex processes simple and wastage of time is avoided. The system can streamline these workflows and thereby minimize confusion and enhance efficiency. It makes sure that everything is done in the correct sequence and in the correct way; it is a smooth experience to the user.

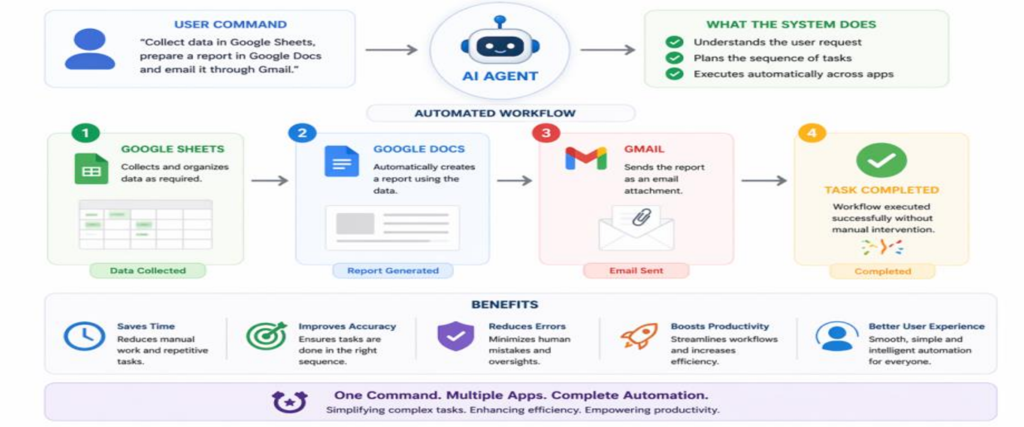

Fig1.3.4.1 Cross-App Workflow

The figure above depicts the operational process of the suggested Unified AI Agent to Automating Tasks across Google Workspace, where the entire end-to-end process of the user request to the completion of a task is presented. It starts with an user giving an order via a natural language interface and then the AI agent interprets it to comprehend the intention and the actions that need to be taken. Also, the figure illustrates the way the AI agent handles the coordination of various applications, making sure that every step is carried out according to the right order and with appropriate data flow.

1.3.5 Scope by Technologies involved.

It is built based on the latest technologies like Artificial Intelligence, Natural Language Processing, and Large Language Models. These technologies integrate to transform the system into a smart system that can perform complicated tasks. Large Language Models give the system reasoning capabilities and the capability to solve multi-step problems. They enable the system to reason logically and strategize on what to do. The combination of these technologies results in the system being powerful, flexible, and able to enhance productivity in real-world situations.

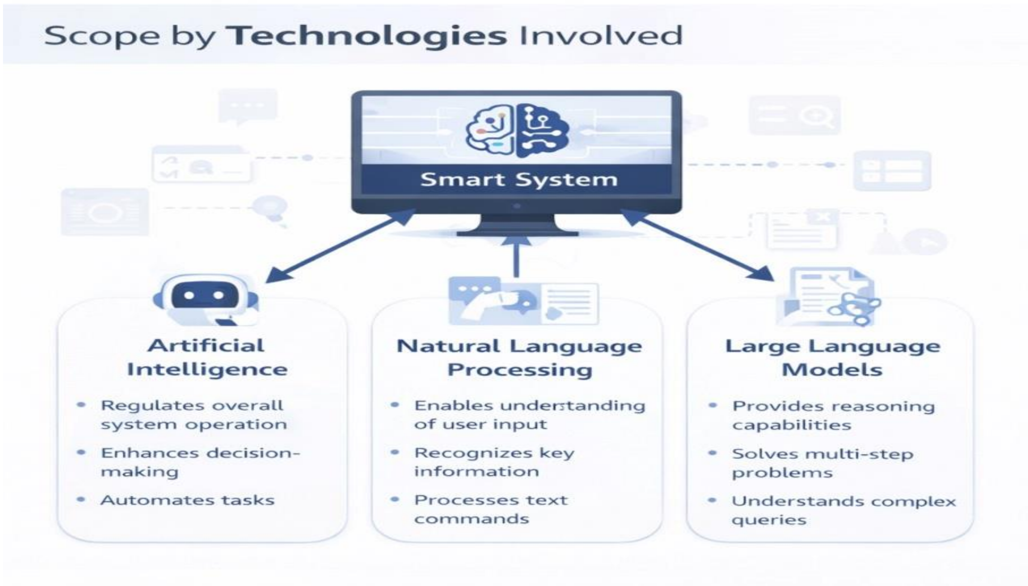

Fig1.3.5.1 System Technologies

The figure above shows the most important technologies employed in the system and the way they collaborate to create an intelligent solution. The core of the diagram is the Smart System, which is the platform of the unified means, and the three primary types of technologies Artificial Intelligence, Natural Language Processing, and Large Language Models are linked to the core. Artificial Intelligence is the central element that determines the work of the entire system, allows automation, making decisions, and performs tasks efficiently.

1.3.6 Scope as per System Functionality.

The system has a systematic procedure, which starts with user input and terminates with completion of tasks. Once a command is given out by a user, the system interprets it to follow what is meant and what should be done. This makes the system have a clear understanding of what is required to be done. Upon understanding the request, the system will plan the steps required and what applications are required. It then implements the tasks with the help of suitable services and APIs. This is done in a step by step manner which makes it possible to handle even complicated tasks in an efficient manner.

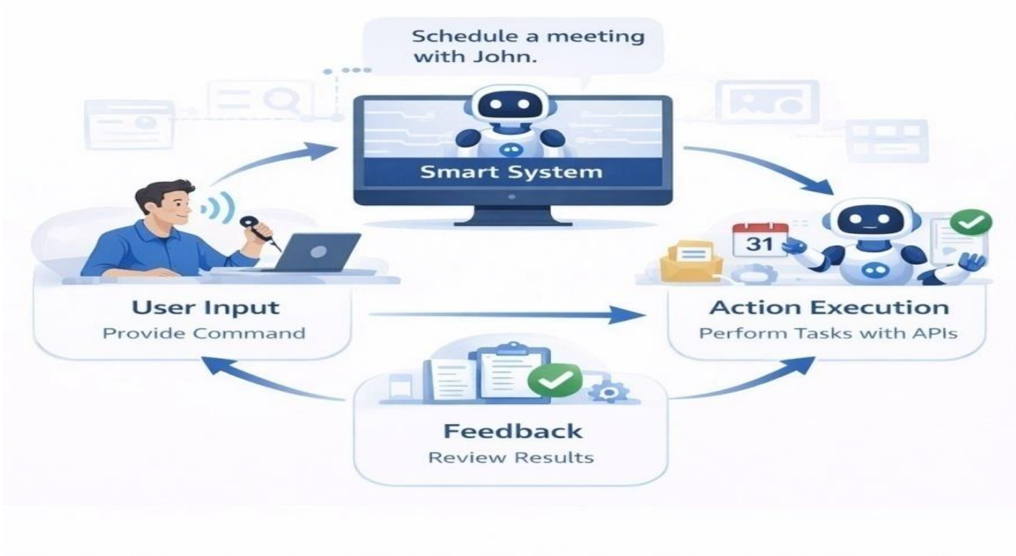

Fig1.3.6.1 System Functionality

The figure above shows how the system works in general, beginning with the input of the user and concluding with the completion of the tasks. It demonstrates the way in which a user makes a command, which is received and interpreted by the smart system. The system interprets the input to know what the user wants and what actions have to be done. This will make sure that the system knows exactly the task that is needed to be done before going any further.

1.3.7 Scope on the basis of Security and Integration.

Important features of the system are security, particularly when user data and multiple applications are involved. The project will also feature a safe integration with Google Workspace services to make sure that all operations are carried out safely. User information is safeguarded by proper authentication means. The system will make sure that the actions performed are only authorized and the data of users is not exposed to unauthorized access. This creates trust and assures the system users that they can trust the system with their day to day activities. The process of integration is also streamlined in order to be reliable. The system has a secure communication with the external services and is able to transmit data safely. This renders the system reliable and fit to the applications in the real world.

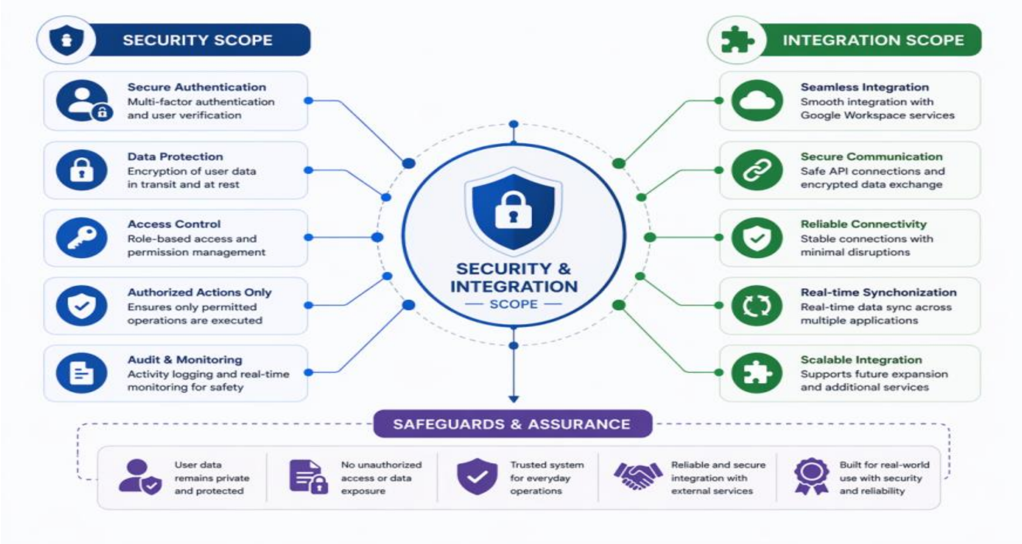

Fig1.3.7.1 Security & Integration

The above diagram is a depiction of the scope of the proposed Unified AI Agent as far security and integration are concerned, and how the system guarantees secure and reliable functioning in a variety of applications. It depicts the use of powerful security measures like user authentication, data encryption and access control to safeguard sensitive user data. All these measures are used to make sure that the system is accessed only by the individuals authorized to access it and that they can only take the necessary actions so that unauthorized access would be impossible, and that data privacy would be preserved. The diagram further shows that the system has secure communication with the Google Workspace services via secure and stable API connections. This would guarantee that all data communications between the AI agent and external applications are carried out in a safe manner without the chances of leakage or data interception.

1.4 Significance of the Project

The suggested Unified AI Agent to Automate Tasks in Google Workspace is significant to enhance productivity and efficiency in contemporary digital workplaces. As the reliance on such tools as Gmail, Google Calendar, Google Drive, Docs, and Sheets rises, users are likely to encounter difficulties with using several applications at the same time. Conventional workflows are manualized, often necessitate application changeover, and repetitive task processing, which minimizes effectiveness. The suggested system will resolve these problems by incorporating several apps into one smart platform that will be driven by Artificial Intelligence. One of the major significances of this system is its ability to improve productivity by automating routine tasks. The email processing, arranging meetings, summarizing documents and sorting files, are automated processes that ease the burden on the users. Using a reduced repetition in the work, the system enables the users to work on more significant and meaningful work, which results in better time management and enhanced efficiency.

The other significant point is that it minimizes the number of manual efforts and mistakes. With the traditional systems, the user manually performs tasks and thus there are high chances of mistakes and inconsistencies. The proposed system is based on smart automation that helps to make sure that tasks are done correctly and on a regular basis. This enhances dependability and minimizes the possibility of mistakes during routine activities. Another benefit of the system over the current tools is seamless integration across applications. Users are able to do tasks on different platforms within one system as opposed to dealing with each application individually. This will remove the necessity to switch between applications regularly and provide a more streamlined workflow, enhancing user experience on the whole. The other important input is the application of Natural Language Interaction, which makes the system user-friendly. The system allows users to communicate with the system through simple and conversational language rather than the use of complex commands. This enables both technical and non-technical users to have access to the system making the curve of learning short and enhancing usability.

Another aspect of the project, which cannot be overlooked, is the significance of smart decision-making with the help of AI technologies. The system is capable of interpreting user intent, processing tasks, and performing them effectively by leveraging Artificial Intelligence, Natural Language Processing (NLP), and Large Language Models (LLMs). This illustrates the way in which the current technologies can be used to address the actual productivity issues. Moreover, the system improves real-time execution of tasks and responsiveness. Tasks are handled in real-time, and users are provided with real-time feedback, which is useful in ensuring the continuity of workflow. This real time feature is necessary in high-speed working situations where there is a need to respond quickly.

2. LITERATURE SURVEY

In recent years, the rapid growth of Artificial Intelligence and cloud-based productivity tools has led to significant advancements in task automation systems. Researchers and developers have focused on building intelligent systems that can reduce manual effort, improve efficiency, and enhance user productivity. The concept of integrating AI with workplace tools such as email systems, document editors, and storage platforms has gained considerable attention due to its practical applications in both personal and professional environments. Early studies in this domain primarily focused on rule-based automation systems, where predefined instructions were used to perform repetitive tasks. While these systems were useful for basic operations, they lacked flexibility and were unable to handle complex workflows involving multiple applications. This limitation led to the development of more advanced approaches using Machine Learning and Natural Language Processing (NLP), enabling systems to understand user commands and automate tasks more intelligently.

With the emergence of Natural Language Processing techniques, researchers began exploring ways to allow users to interact with systems using human language. Models such as BERT and GPT have significantly improved the ability of machines to understand context, intent, and meaning in user inputs. This advancement made it possible to design systems that can interpret commands like sending emails, creating documents, or scheduling meetings without requiring manual navigation through multiple interfaces. Another important area of research is the integration of multiple applications through APIs. Several studies have highlighted the importance of seamless communication between services such as Google Workspace, Microsoft Office, and cloud storage platforms. API-based integration allows systems to connect different tools and perform tasks across them in a coordinated manner. However, many existing systems focus on single-application automation and do not fully utilize cross-application workflows, which limits their effectiveness in real-world scenarios.

Recent research has also explored the concept of intelligent agents and multi-agent systems, where AI agents can perform tasks autonomously and collaborate with different services. These systems are capable of breaking down complex tasks into smaller steps and executing them in sequence. For example, generating a report from data and sending it via email can be handled automatically by an intelligent agent.

Despite these advancements, many existing solutions still face challenges related to scalability, integration complexity, and real-time performance.

Security and privacy have also been key concerns in the development of automation systems. Studies emphasize the importance of secure authentication mechanisms, data encryption, and access control to protect user information. As systems interact with multiple applications and handle sensitive data, ensuring safe communication and preventing unauthorized access becomes critical.

S.No	Title of the Paper	Author of the Paper	Methodology Used	Drawbacks of their Methodology	Research Gap Identified
1	Intelligent Email Automation using NLP	Kumar et al.	NLP-based email classification	Limited to email tasks only	No multi-application automation
2	AI-Based Task Scheduling System	Sharma et al.	Machine Learning for scheduling	No integration with external apps	Lack of cross-platform workflow
3	Conversational AI for Productivity Tools	Lee et al.	NLP + Chatbot models	Limited contextual understanding	No deep task automation
4	Workflow Automation using APIs	Johnson et al.	API-based integration	Complex implementation	No intelligent decision-making
5	Smart Document Generation using AI	Patel et al.	NLP + Template-based generation	Limited flexibility	No integration with other services
6	Multi-Agent Systems for Task Automation	Wang et al.	AI agents collaboration	High system complexity	Limited scalability
7	Cloud-Based Automation Systems	Gupta et al.	Cloud computing + APIs	Security concerns	Weak authentication mechanisms

8	AI Assistants for Office Productivity	Brown et al.	Virtual assistant models	Limited to single platform	No unified system approach
9	Task Automation using RPA (Robotic Process Automation)	Singh et al.	Rule-based automation	Not adaptive or intelligent	No learning capability

Table 2.1 Literature Survey

Furthermore, modern research highlights the need for user-centric design, focusing on improving user experience through intuitive interfaces and minimal interaction effort. Systems that are easy to use and require less technical knowledge are more widely adopted. Building on these developments, recent trends are increasingly focused on creating more context-aware and adaptive automation systems. Modern AI-driven platforms are now capable of learning from user behavior over time, allowing them to provide personalized automation experiences. For instance, systems can analyze patterns in user activities—such as frequently scheduled meetings or repeated email responses—and proactively suggest or execute actions without explicit commands. This shift from reactive to proactive automation significantly enhances productivity and reduces cognitive load on users.

Another emerging direction is the use of large language models (LLMs) integrated with real-time data sources, enabling dynamic decision-making. These systems can not only understand user intent but also retrieve relevant information, generate content, and execute multi-step workflows seamlessly. Additionally, advancements in cloud computing and microservices architecture have improved the scalability and deployment of such systems, making them more accessible and efficient. However, challenges remain in ensuring reliability and minimizing errors in automated decision-making. There is also a need for better user control and transparency, so users can understand and override system actions when necessary.

3. DESIGN

3.1 System Architecture

The proposed Unified AI Agent to Automating Tasks across Google Workspace is organized in a modular and layered system architecture to guarantee an efficient processing, scalability and smooth interaction between various components. The architecture aims at turning the input of users into meaningful actions by combining several applications of the Google Workspace using intelligent automation. All the layers in the system have a given role to play and combined they result in a seamless flow of work input to output. The architecture follows logical flow in which the user engages with the system via a user-friendly interface, and the system executes the request by applying sophisticated systems technology like Natural Language Processing (NLP) and Artificial Intelligence (AI).

The system can interpret user will, organize tasks and perform them across a variety of applications such as Gmail, Calendar, Drive, Docs and Sheets. Such a multi-layered approach will make even a multi-step task to be effectively managed.

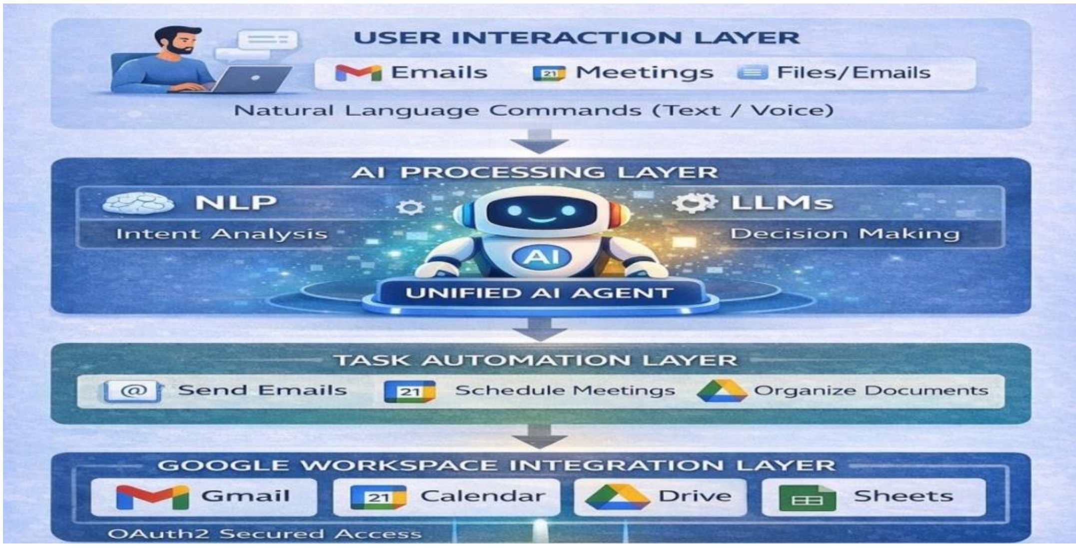

Fig3.1.1 System Architecture

The following figure depicts the stratified design of the proposed Unified AI Agent system, where user input is converted to automated processes. The User Interaction Layer is placed on the top whereby users can give commands using the natural language either via text or voice such as handling emails, scheduling meetings or handling files. This information enters into the AI Processing Layer, where the user intent is studied and smart decisions are made by technologies such as Natural Language Processing (NLP) and Large Language Models (LLMs). This layer becomes the heart of the system as it allows the system to comprehend requests and strategize accordingly.

3.1.1 User Interaction Layer

The User Interaction Layer will serve to provide the point of entry into the system where users interact with the AI agent. This layer gives anyone a simple and easy to understand interface where he/she can use natural language to give commands. The interaction may be either text or voice based and this makes the system to be accessible to a large user population. The primary purpose of this layer is to make it easy to use, and to have a pleasant experience of interaction. Users will not have to use various applications or master intricate interfaces. Rather, they are able to just articulate their needs in a conversational tone and the system does the rest.

Also, this layer will be tasked with the responsibility of presenting outputs like confirmations, generated content and notifications. It makes the system more user-friendly and efficient as it guarantees that users obtain understandable and clear answers.

3.1.2 Natural Language Understanding Layer

The Natural Language Understanding (NLU) Layer has the task of understanding what the user can input and transforming it into a structured information that can be processed by the system. This layer implements Natural Language Processing to classify the intent of a user and to recognize significant information like dates, names, email addresses, document snippets.

The layer is important in sealing the gap between human language and machine interpretation. It makes sure that the system gets the user just what he wants, with either a complex or conversational input. This enhances precision in performance of tasks. Moreover, NLU layer makes the system more responsive to various types of input and will accept various user commands. It allows the system to react in an intelligent manner and take the necessary actions to react to its environment and intent.

3.1.3 AI Reasoning and Task Planning Layer

The brain of the system is the AI Reasoning and Task Planning Layer. This layer then interprets the user input and then evaluates the request and finds the order of actions needed to accomplish the task. It applies AI models and logic to divide more complex tasks into smaller manageable tasks.

This layer can implement multi-step workflows, which require the usage of multiple applications. An example is when a user requests to create a report and email it, the system will take steps of retrieving data, creating a document and mailing the document. This makes sure that tasks are carried out in the proper sequence. Moreover, the layer can facilitate intelligent decision making by choosing the right services and actions that are needed in every task. It makes sure that the system is efficient and provides the correct results.

3.1.4 Task Execution Layer

Task execution layer is tasked with executing the actions that are scheduled by AI reasoning layer. This layer comprises special modules which communicate with various Google Workspace services to carry out certain operations. Each of the modules is created to process a specific kind of task, e.g. email management, calendar scheduling, document processing or data handling. These modules translate the actions planned into API calls and implement them in the respective applications.

This layer makes sure that tasks are accomplished in an accurate and efficient manner. This is because the system is flexible in that by dividing the execution into various modules, the system is capable of executing a number of tasks at the same time without any conflict.

3.1.5 Integration Layer

The Integration Layer is a system that will connect the AI system to the Google Workspace applications. Using APIs, it allows communicating with external services (Gmail, Google Calendar, Google Drive, Google Docs, and Google Sheets) via the system. This layer guarantees safe and trustworthy communication with external services. It handles authentication and authorization and enables the system to act on behalf of the user and keep data privacy and security.

Moreover, the integration layer aids in smooth transfer of data among applications. This enables the system to execute cross-application workflows effectively such that complex tasks that require many services can be automated.

3.2 UML Diagrams

3.2.1 Use Case Diagram

The Use Case Diagram can be used to give a general picture of the interactions between various users and the Unified AI Agent to Automating Tasks in Google Workspace. It aids in the functionalities of the system and roles of various users. The diagram illustrates the association between users (actors) and the workings of the system, and thus it is simpler to see how the system is utilized in real-life situations.

Three main participants are involved in this system: User, Admin, and System Services (Google Workspace APIs). Main actor is the User who communicates with the AI agent to accomplish activities like emailing, meeting scheduling, document writing, and file management. The user enters commands to the system in natural language and gets responses in form of confirmations, generated text or notifications. The Admin is instrumental in the administration of the system. The duties of the admin include keeping track of the performance of the system, integrations, error handling and making sure that everything is working right. The admin also manages updates of systems and ensures data security.

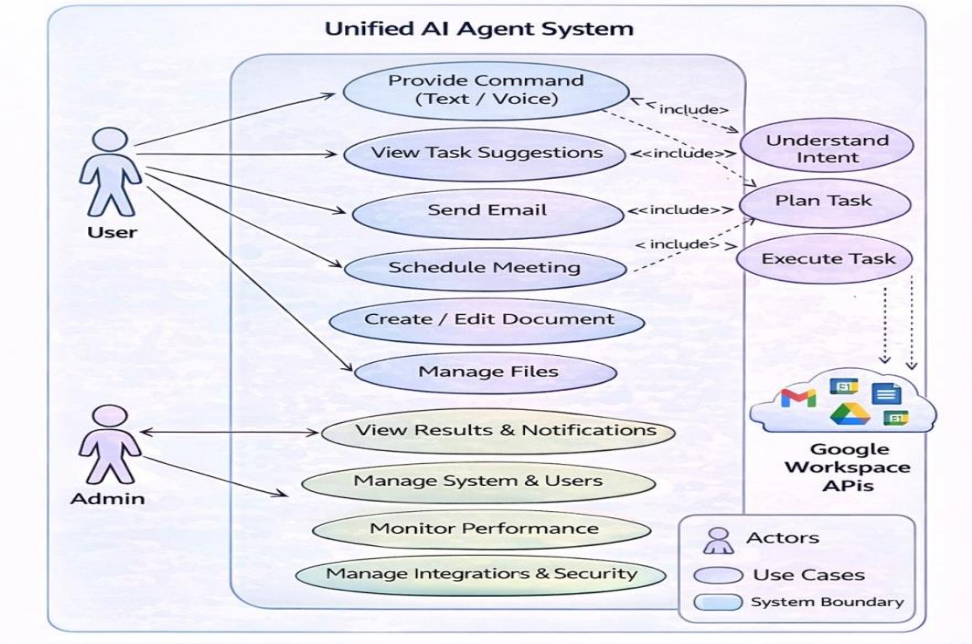

Fig3.2.1.1 Use Case Diagram of Unified AI Agent System

The use case diagram represents how different users interact with the Unified AI Agent system and highlights the main functionalities provided by the system. It shows three key actors: the User, Admin, and Google Workspace APIs. The user interacts with the system by giving commands such as sending emails, scheduling meetings, creating documents, and managing files. These actions are processed by the system through internal steps like understanding intent, planning tasks, and executing them.

The diagram also illustrates the role of the admin, who is responsible for managing system operations such as monitoring performance, handling users, and maintaining integrations. The inclusion relationships between use cases show that tasks like sending emails or scheduling meetings depend on core processes like understanding intent and task execution.

3.2.2 Sequence Diagram

The Sequence Diagram shows the chronological sequence of interactions of various system components. It displays the way in which a user request is handled starting with the first request and ending with the final response, and shows the order of operations that the system runs. The process starts with the User giving a command to the system e.g. asking the system to send an email or to schedule a meeting. The User Interface receives this request and sends it to Natural Language Processing module to be interpreted.

The NLP module in the following step analyses the input and identifies the important information about the input including intent, entities and context. This processed information is then forwarded to the AI Reasoning and Task Planning module that decides on the course of action to accomplish the task. In case the task is complicated, then it is further subdivided into several sub tasks. After planning, the request will be sent to the Task Execution Layer, which will then be sent to various modules that will do certain tasks. The email module communicates with Gmail APIs and the calendar module communicates with Google Calendar APIs, to give an example. These modules perform tasks either in parallel or in a sequence depending on the needs.

This diagram can demonstrate the flow of data within the system in a clear manner and the interaction between various components within the system. It emphasises how effective the system is to manage tasks and how all the operations are to be carried out in a rational order.

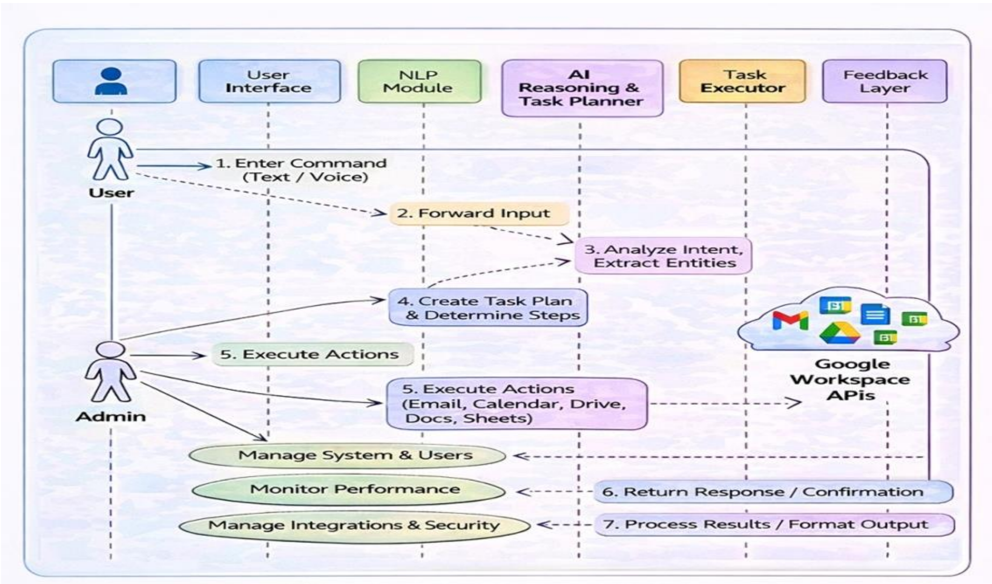

Fig3.2.2.1 Sequence Diagram of Unified AI Agent System

The sequence diagram explains how different components of the system interact with each other over time when a user gives a command. The process begins when the user enters a command through the interface. This input is passed to the NLP module, where the system analyzes the request and extracts important information such as intent and entities. The processed information is then sent to the AI reasoning and task planning module.

After planning the task, the system forwards the instructions to the task executor, which performs actions using Google Workspace APIs like Gmail, Calendar, or Drive. Once the task is completed, the system sends the result back to the feedback layer, which displays the output or confirmation to the user. This diagram clearly shows the step-by-step communication between components and ensures that all operations are executed in a logical sequence.

3.2.3 Activity Diagram

Activity Diagram is used to depict the workflow of the system, which indicates the way various activities are carried out in different stages. It gives a clear picture of the overall process, both in regards to inputting the user input and the final product. Start node marks the beginning of the workflow, that is, the process starts. The initial activity is the Receive User Input in which the system receives the commands by the user. This is then succeeded by the Process Input stage where NLP processes are used to get to know the request.

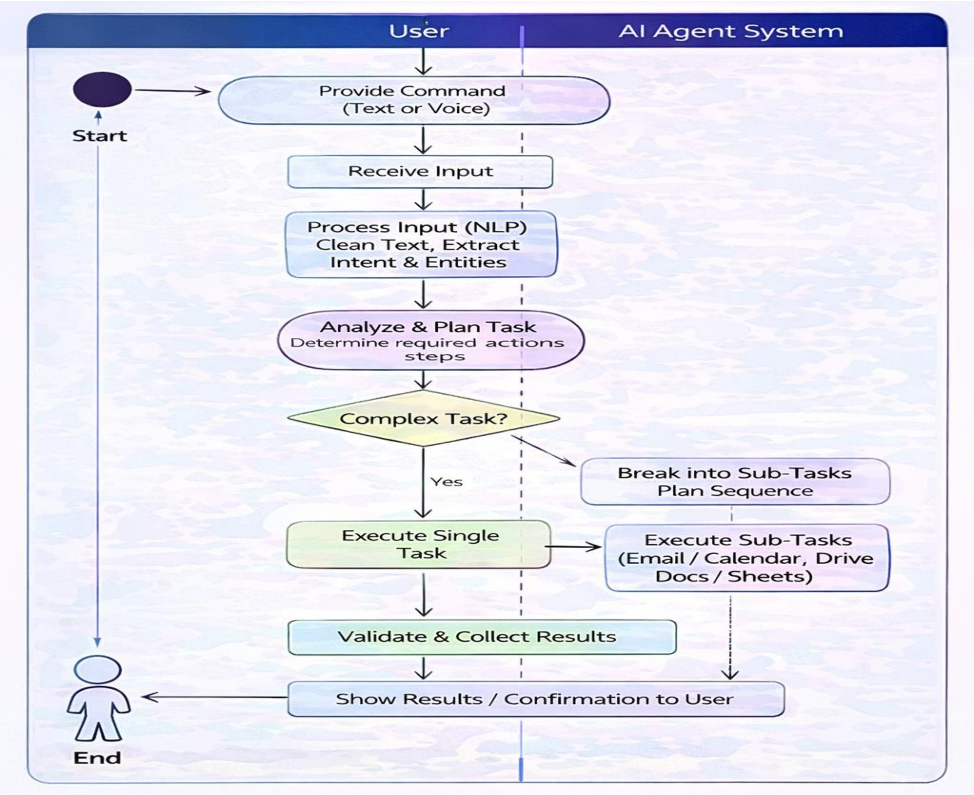

Fig3.2.3.1 Activity Diagram of Unified AI Agent System

The activity diagram represents the workflow of the system from start to end, showing how a user request is processed and executed. The process begins when the user provides a command in text or voice format. The system then receives the input and processes it using Natural Language Processing to extract intent and relevant details. Based on this understanding, the system analyzes the request and plans the required actions.

A decision point is included to determine whether the task is simple or complex. If the task is simple, it is executed directly; otherwise, it is broken into smaller sub-tasks and executed step by step. After execution, the system validates the results and generates the final output. The workflow ends by displaying the results or confirmation to the user. This diagram helps in understanding how the system handles tasks efficiently and ensures smooth execution.

3.2.4 Class Diagram

The Class Diagram is a way to describe the structure of the system and it displays various classes, their attributes, methods and relationships. It offers a fixed perspective of the system and assists in the comprehension of the manner in which components are structured. The core classes in the system are as follows: User, Admin, Command Processor, NLP Module, Task Planner, Task Executor, Integration Manager and Notification System. The system has a role assigned to each of the classes. The User class is the end users and has characteristics like the name and the user id. It has such methods as sendCommand() and receiveResponse. In the Admin class, there are functions like manageSystem() and monitorPerformance, and it is possible to control system activities.

The associations of these classes demonstrate the direction of the flow of data in the system. The Command Processor communicates with the NLP Module and Task Planner with the User. The Task Executor is used to execute actions and the Notification System is used to give feedback.

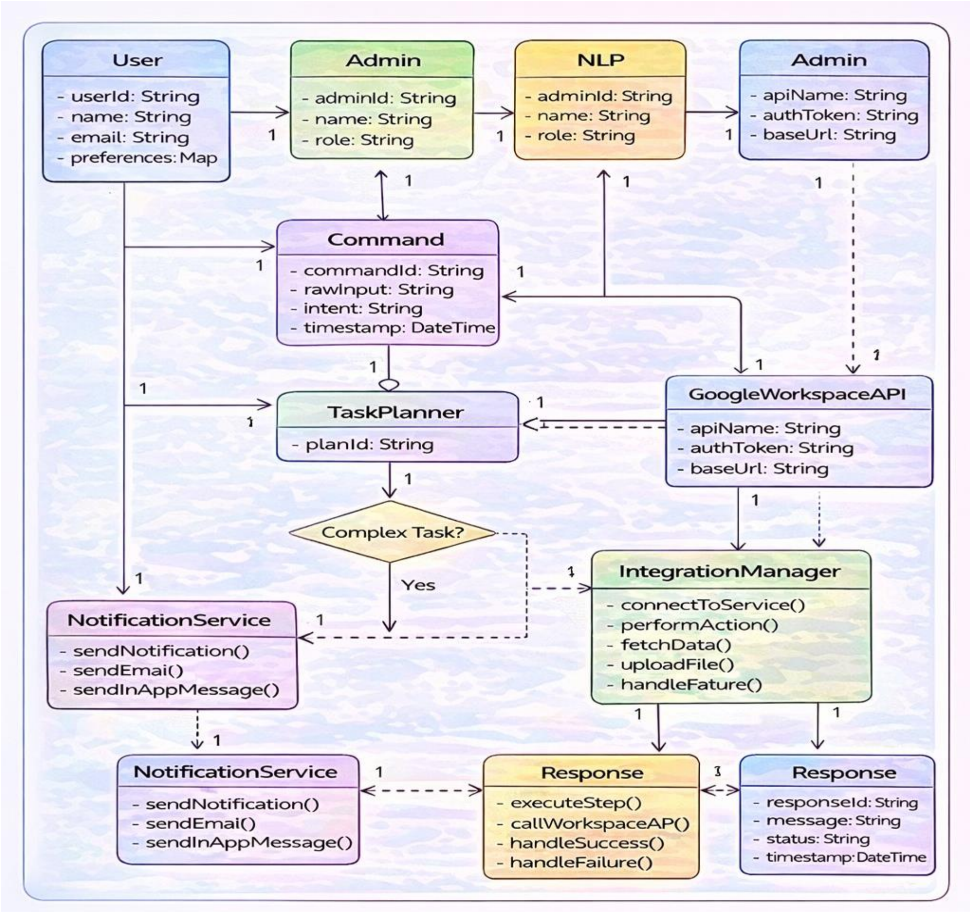

Fig3.2.4.1 Class Diagram of Unified AI Agent System

The class diagram shows the structural design of the system by representing different classes, their attributes, methods, and relationships. It includes important classes such as User, Admin, Command, NLP Module, Task Planner, Task Executor, Integration Manager, and Response. Each class has a specific role in the system. For example, the User class handles interactions, while the NLP module processes input and extracts meaning. The relationships between these classes illustrate how data flows within the system. The command is processed by the NLP module, planned by the task planner, and executed by the task executor using integration services. The response class manages the output, and the notification service ensures communication with the user.

4. METHODOLOGY

The proposed Unified AI Agent to Automate Tasks across Google Workspace has a methodology that explains the process of designing, developing, and implementing the system. It explains the combination of various technologies, tools and processes to create a smart system of automation that is able to perform tasks in a real world effectively. The system is aimed at streamlining interactions with the user via translating natural language instructions into meaningful activities within different applications. This is unlike the traditional systems that require manual navigation, and independent handling of applications but the system adopts an integrated approach where all the operations are handled by a single intelligent agent. This methodology is based on the application of Artificial Intelligence, Natural Language Processing, and smart planning of tasks to automate the workflow and enhance productivity. The overall methodology encompasses various steps like interpreting user input, language processing, task planning, action execution, integration with external services and feedback. The stages are important in making sure that the system functions effectively and provides the correct results.

4.1 Methods

The system combines various contemporary methods of computations to deal with various elements of automating tasks. Each approach is set to address a particular issue, i.e. to comprehend user input, to make a decision, or to perform actions. Efficiency and reliability of the system are guaranteed by the combination of these methods.

4.1.1 Natural Language Processing (NLP)

One of the most significant parts of the system is the Natural Language Processing, as the system can comprehend the user commands in a plain language. The system enables users to talk in a natural and conversational manner, as opposed to having to put structured inputs into the system. This will make the system more accessible and user-friendly.

The NLP process starts by the analysis of the input text in order to clean and prepare it to be fed into the rest of the process. This involves elimination of unwarranted symbols, splitting the text into bits, and isolating meaningful words. The system then digs out valuable information like intent, keywords, dates and names. Considering a command such as Schedule a meeting tomorrow at 10 AM, the system recognizes the action (schedule), the time (10 AM) and date (tomorrow).

Furthermore, the system employs the techniques of intent recognition in order to comprehend what the user is supposed to accomplish. This enables it to distinguish various kinds of requests like sending emails, making documents or arranging files. NLP will provide the user with the correct response to the input by interpreting user input accurately to a response and do the required task.

4.1.2 Artificial Intelligence and Reasoning

The main part of the system is Artificial Intelligence as the decision-making part. After the user input has been processed, the AI module will analyze the request and decide how to perform the task in the most appropriate manner. This entails critical thinking and strategizing using the information at hand. Complex tasks are broken down into smaller steps with intelligent algorithms used in the system. To illustrate, when a user requests to prepare a report and send it via email, the system recognizes several processes including: data collection, document generation and emailing. The system is more powerful and flexible due to this capability to support multi-step tasks.

The reasoning part is also used to decide on the suitable services and actions that should be taken to each task. It makes sure that the tasks are performed in the right sequence and error-free. This enhances efficiency and minimizes the possibility of failure.

4.1.3 Planning and Workflow Management of tasks

One of the key steps in the system is task planning, during which the AI agent plans the order of actions to execute a request. The system has a workflow to perform its tasks in an orderly manner, rather than performing them randomly in order to guarantee accuracy and efficiency. Planning is done by recognizing the dependencies of tasks and sequencing them in the right sequence. As an illustration, prior to dispatching an email that contains an attachment, the system has to initially create or access the document. This rational order provides that work is accomplished successfully.

The system is also capable of dynamic workflow management, i.e., it can modify the plan according to the evolving conditions or other inputs. In case of failure or any changes, the system is able to correct the workflow or update it. This will enhance reliability and make sure that even complicated activities can be dealt with easily without human interference.

4.1.4 API Integration and Task-execution

Tasks are executed by integrating with Google Workspace services by using APIs. APIs are a communication link between the system and external applications, which enable the system to carry out activities like emailing, holding of meetings, and file management. Every activity is transformed into API calls, which are implemented in the respective application. An example of this is an email message where one interacts with the Gmail API and a meeting where it is the Google Calendar API. This will make sure that tasks are carried out directly on the platform required. The system employs secure authentication to provide secure access to user data. It can also process API requests effectively to accommodate several tasks without delays. Through API-based execution, the system is accurate, reliable, and can be used in real time in automating tasks.

4.1.5 Automation and Smart execution.

The most important goal of the system is automation, i.e. the repetitive tasks are carried out automatically without assistance of the user. The system will be in place to deal with the routine tasks, like email management, scheduling and processing documents effectively. After planning a task, the system automatically executes the task by coordinating the various modules and services. This minimizes working manually and makes work faster. The system also brings about some consistency in execution adhering to some logic and smart processing. This reduces errors and enhances reliability. Automation is time saving as well as productivity boosting since users have time to concentrate on other more valuable tasks.

4.2 Tools

The creation of the offered Unified AI Agent to Automate tasks in Google Workspace implies that an extensive set of contemporary tools, frameworks, and platforms are utilized. These tools assist various phases of the system such as input processing, intelligent decision making, execution of tasks, integration and visualization. All tools are chosen because they are capable of enhancing efficiency, flexibility, and performance of the entire system. As the fundamental programming language, Python is employed at the heart of the system because of its simplicity, readability, and high level of support of Artificial Intelligence and automation. Python also offers an ecosystem of libraries that enable it to be easy to use in Natural Language Processing, API integration, and automated intelligent work. Its scalability enables developers to develop maintainable and scalable systems easily.

Large Language Models (LLMs) and AI-based systems are also used in the system to reason and plan tasks. These models can assist the system to make sense of the complicated requests, subdivide them into smaller operations and come up with smart answers. This greatly enhances the system to deal with multi-step work flows and real world situations.

Environments, like Jupyter Notebook and Google Colab are used to develop and test. Moreover, the integrated development environments (IDEs) are VS Code and Python Charm, which are utilized in writing, managing and structuring code effectively. Google Workspace APIs, such as Gmail API, Google Calendar API, Google Drive API and Google Docs API are also essential in the system. These APIs allow the system to communicate with third party services and allows it to carry out activities like emailing, creating meetings and handling files. OAuth 2.0 which provides safe and reliable access is used to secure authentication.

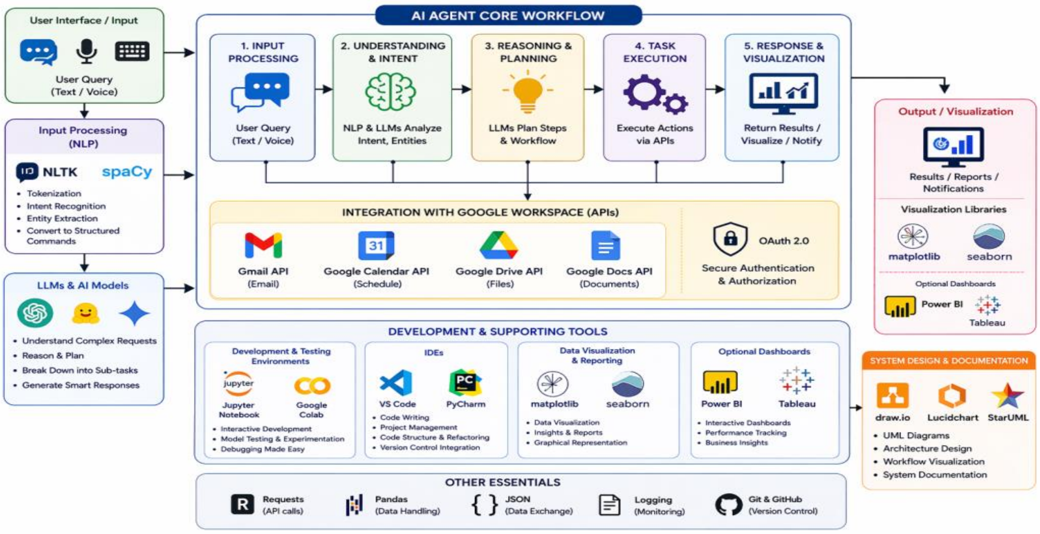

Fig4.2.1 Tools Architecture

This diagram represents the architecture of a Unified AI Agent designed to automate tasks within the Google Workspace ecosystem. It shows a clear workflow starting from user input (text or voice), which is processed using Natural Language Processing tools like NLTK and spaCy. The system then uses AI models and reasoning engines (LLMs) to understand the request, break it into steps, and plan actions. These actions are executed through integration with Google Workspace APIs such as Gmail, Google Calendar, Google Drive, and Google Docs, enabling automation of tasks like sending emails, scheduling meetings, and managing files. Secure access is handled using OAuth 2.0.

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4.3 Materials

The resources involved in the design of the proposed system will comprise hardware and software (both) as well as sources of data and interfaces with the system. These resources will be needed to install, test and deploy the system to a real world setting. They make sure that the system is effective and provides dependable output. As far as hardware is concerned, the system must be capable of supporting a computer or laptop with enough processing power to support multiple tasks at an exceptionally given time. It is also recommended to have at least 8GB RAM to provide a smooth performance and to work with AI models and to handle several API requests. It also requires a stable internet connection since the system relies on the cloud-based services and real-time data exchange. Besides the minimal hardware, an optional yet advantageous feature is the use of a GPU (Graphics Processing Unit

Software-wise, the system needs a Python-based system with all the required libraries and frameworks installed. These include NLP libraries, API integration tools, and AI frameworks. Depending on the preference of the user and compatibility with the system, the operating system may be Windows, Linux or macOS. The system also uses data inputs given by the users that comprises of natural language commands like scheduling meetings, e-mailing or document generation. This system is more dynamic and interactive as compared to traditional systems which rely on structured datasets, and therefore, the user input is mostly real-time.

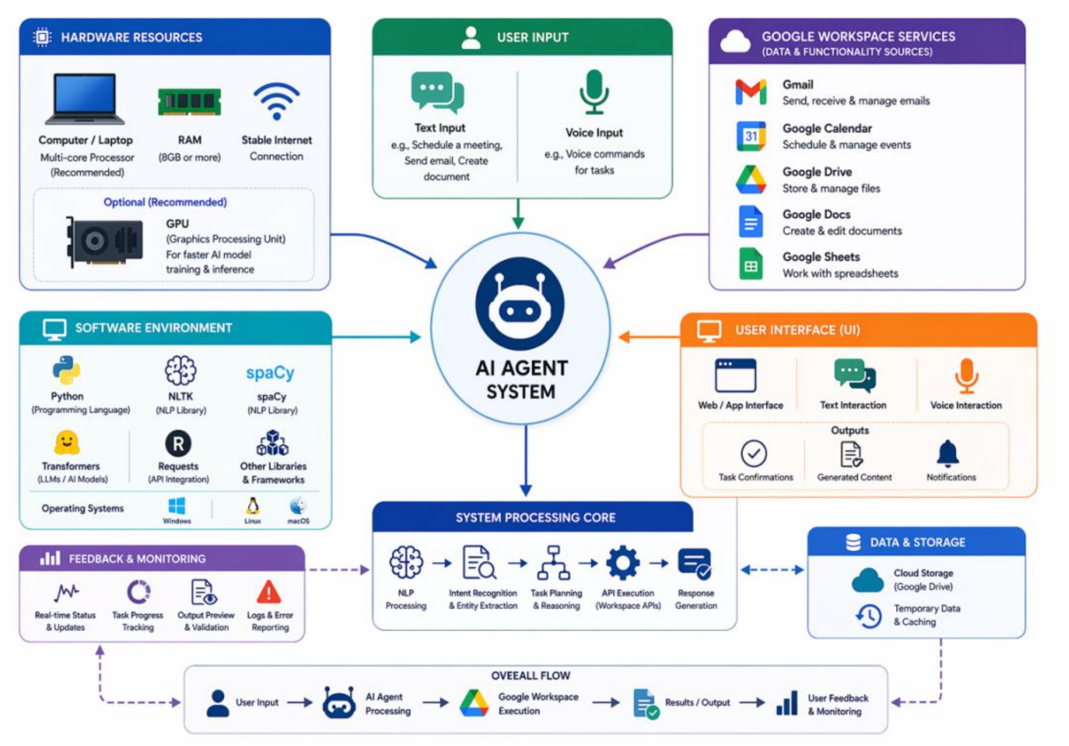


Fig4.3.1 System Resources

This diagram illustrates the resource architecture of a Unified AI Agent system designed to automate tasks within the Google Workspace environment. At the center is the AI Agent System, which acts as the core processing unit. It receives user input in the form of text or voice commands and processes it using NLP and AI models to understand intent and plan actions. On the left, the Hardware Resources section shows the physical requirements such as a computer or laptop, sufficient RAM (8GB or more), stable internet connection, and an optional GPU for faster processing of complex AI tasks.

4.4 Data Sources, Storage, and Security

4.4.1 Data Sources from Google Workspace

The Unified AI Agent utilizes data from various Google Workspace applications as its primary source of information to perform intelligent task automation. These data sources include Gmail, Google Docs, Google Sheets, Google Drive, and Google Calendar, each contributing different types of structured and unstructured data. Emails from Gmail provide communication data such as messages, attachments, and user interactions. Google Docs and Sheets offer textual and numerical data used for document generation, reporting, and analysis. Google Calendar provides scheduling data such as events, reminders, and meeting details, while Google Drive acts as a centralized repository for file storage and access. The system retrieves this data securely using Google APIs, ensuring real-time access and synchronization across applications. These datasets are essential for enabling cross-application workflows, where tasks involve multiple services simultaneously.

For example, creating a report may involve retrieving data from Sheets, generating content in Docs, and storing it in Drive. The diversity of data sources allows the Unified AI Agent to understand user activities comprehensively and execute tasks efficiently. Overall, Google Workspace data forms the backbone of the system, enabling seamless automation and integration across different productivity tools.

4.4.2 Data Storage and Management

The Efficient data storage and management are critical components of the Unified AI Agent to ensure smooth operation and fast task execution. The system stores data in structured formats such as JSON, relational databases, or cloud-based storage systems, depending on the type and usage of the data. Temporary data, including user session information and intermediate task outputs, is stored in cache systems to enable quick access and reduce processing time. Persistent storage is used for maintaining logs, user preferences, and workflow history, which helps the system learn and improve over time. Cloud-based storage solutions are preferred due to their scalability, flexibility, and real-time synchronization capabilities. These systems allow the agent to handle large volumes of data without performance degradation. Data preprocessing techniques such as cleaning, normalization, and formatting are applied before storage to ensure consistency and accuracy. Additionally, indexing and efficient querying mechanisms are implemented to enable fast retrieval of information during task execution.

4.4.3 Data Security and Privacy

Data security and privacy are essential considerations in the design of the Unified AI Agent, as it handles sensitive user information across multiple Google Workspace applications. The system implements secure authentication mechanisms such as OAuth 2.0 to ensure that only authorized users can access their data. Access tokens are used to interact with Google APIs securely, and permissions are strictly controlled based on user consent.All data transmitted between the system and external services is encrypted using secure communication protocols to prevent unauthorized access and data breaches. The system also enforces role-based access control, ensuring that users can only perform actions permitted within their scope.

In addition, the system follows best practices for data privacy, including minimal data retention and secure storage of credentials. Regular monitoring and logging mechanisms are implemented to detect and prevent suspicious activities. By prioritizing security and privacy, the Unified AI Agent builds user trust while ensuring safe and reliable automation across multiple platforms.

4.5 Workflow

The proposal system workflow is the entire process of converting a request made by a user into an automated process. It outlines the way in which the system takes the input, intelligently processes it, performs tasks and delivers results. Such an organized workflow will guarantee the flow of operations and proper fulfillment of tasks.

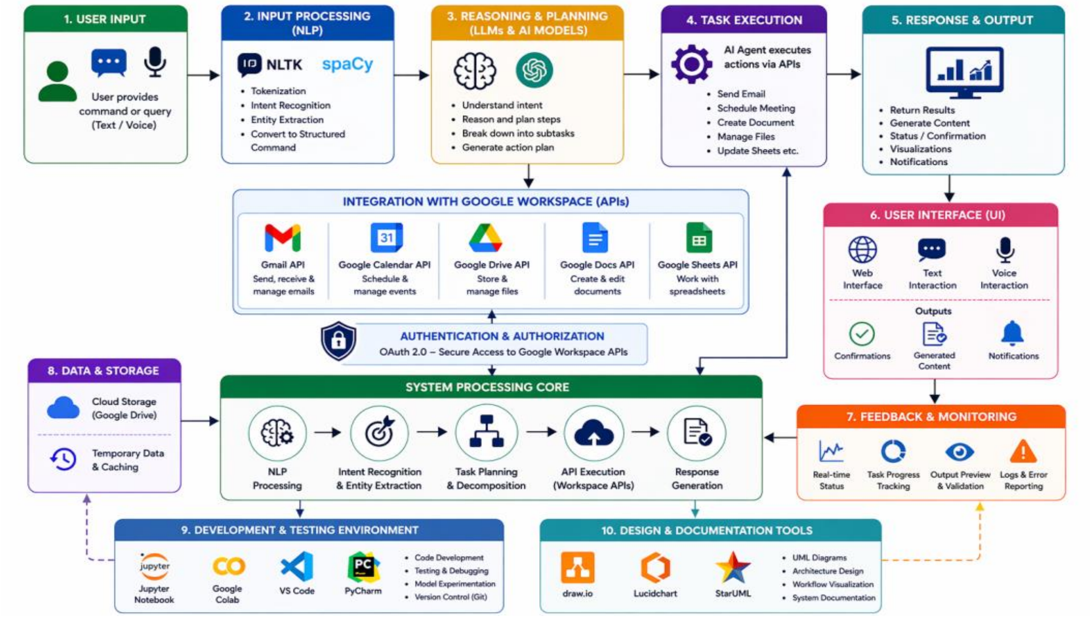


Fig4.5.1 Workflow Diagram

4.5.1 User Input

Workflow starts with the user giving a command to the system. This input can be provided either in the form of text or voice in natural language and interaction becomes easy and natural. The system will be developed to take the flexible input and not in a strict format and enables the user to communicate normally. This phase is significant as it is the point of departure of the whole process.

4.5.2 Processing and Understanding of inputs

After receiving the input, it is sent to Natural Language Processing module. At this phase the system examines the input to find out what the user wants. It recognizes such significant factors as actions, time, names and references. The system also eliminates the irrelevant words and concentrates on the relevant information. This makes sure that the command is translated into structured format which can be further processed.

4.5.3 Task Planning

The system proceeds to the planning stage after comprehending the input. In this case, the AI element defines the steps to follow in the accomplishment of the task. In case the request is complicated, it is broken down into minor steps. As an illustration, a request such as the one that involves preparation of a document and emailing it, entails several procedures including the creation of the document, its saving and attaching it to an email.

4.5.4 Task Execution

The system carries out the intended actions at this stage as it interacts with Google Workspace services. All the actions are translated into API calls and implemented into the corresponding application. As an illustration, the system will utilize Gmail API to send emails, Google Calendar API to organize meetings, and Google Drive API to access files. This enables the system to undertake tasks directly in the requisite platform.

4.5.5 Integration and Coordination

In the execution process, the system will maintain a good coordination among various applications. When there are several services in a task, the system automatically deals with the flow of data among the services. An example is that a document prepared in Google Docs can be automatically sent as an email in Gmail without having to do so manually. This integration saves on complexity and enhances the efficiency of workflow. This step accentuates the fact that the system can deal with cross-application workflows in an efficient manner.

5. IMPLEMENTATION

App.py

```
from flask import Flask, render_template, request, jsonify, session, redirect, url_for, send_file
import threading
from dotenv import load_dotenv
from google_auth_oauthlib.flow import Flow
from google.oauth2.credentials import Credentials
import requests
import json
import io
from flask_socketio import SocketIO, emit
import base64
from datetime import datetime, timedelta
from bson import ObjectId
from bson.json_util import dumps
import re

# Agent and Analytics imports
from handlers.agent_orchestrator import AgentOrchestrator
from models.agent_model import AgentModel
from handlers.analytics import AnalyticsHandler
from handlers.scheduler import SmartScheduler
app = Flask(_name_)
app.secret_key = secrets.token_hex(16)
app.config['MAX_CONTENT_LENGTH'] = 16 * 1024 * 1024  # 16MB max
app.config['DEBUG'] = True
app.config['SESSION_TYPE'] = 'filesystem'
```

```
# ===== MONGODB CONFIGURATION =====
app.config["MONGO_URI"] = os.getenv("MONGO_URI", "mongodb://localhost:27017/workspace_agent")
mongo = PyMongo(app)

# Initialize collections and models
friends_collection = None
history_collection = None
friend_model = None
history_model = None
agent_model = None
agent_orchestrator = None
analytics_handler = None
scheduler = None

try:
    mongo.db.command('ping')
    friends_collection = mongo.db.friends
    try:
        friends_collection.create_index([("user_id", 1), ("name", 1)], unique=True)
        friends_collection.create_index([("user_id", 1), ("email", 1)])
        print("✅Friends collection indexes created")
    except Exception as e:
        print(f"□ Friends index creation warning: {e}")
    history_collection = mongo.db.history
    try:
        history_collection.create_index([("user_id", 1), ("timestamp", -1)])
        history_collection.create_index([("user_id", 1), ("action", 1)])
        print("✅History collection indexes created")
```

```
    except Exception as e:
        print(f"□ History index creation warning: {e}")

    # Initialize agents collection
    agents_collection = mongo.db.agents
    try:
        agents_collection.create_index([("user_id", 1), ("created_at", -1)])
        agents_collection.create_index([("user_id", 1), ("status", 1)])
        print("✅Agents collection indexes created")
    except Exception as e:
        print(f"□ Agents index creation warning: {e}")
```

Frontend/app.js

```
import os
import re
import base64
from googleapiclient.discovery import build
from googleapiclient.http import MediaIoBaseDownload
from google.oauth2.credentials import Credentials
from email.mime.multipart import MIMEMultipart
from email.mime.base import MIMEBase
from email.mime.text import MIMEText
from email import encoders

# Optional parsers
try:
    import PyPDF2
except ImportError:
    PyPDF2 = None
```

```
try:
    import docx
except ImportError:
    docx = None
try:
    from pptx import Presentation
except ImportError:
    Presentation = None

# Load environment variables
load_dotenv()
cohere_api_key = os.getenv("COHERE_API_KEY")
if not cohere_api_key:
    raise ValueError("❌COHERE_API_KEY not found in .env file")

co = cohere.Client(cohere_api_key)

SCOPES = [
    "https://www.googleapis.com/auth/drive.readonly",
    "https://www.googleapis.com/auth/spreadsheets.readonly",
    "https://www.googleapis.com/auth/gmail.send",
    "https://www.googleapis.com/auth/documents.readonly",
    "https://www.googleapis.com/auth/presentations.readonly"
]
```

Backend/app.py

```
import os
import pickle
import pathlib
```

```
from google.auth.transport.requests import Request
from google.oauth2.credentials import Credentials
from google_auth_oauthlib.flow import InstalledAppFlow

# Define all the scopes you need for the complete Workspace Agent
SCOPES = [
    # Drive scopes
    "https://www.googleapis.com/auth/drive.readonly",
    "https://www.googleapis.com/auth/documents.readonly",
    "https://www.googleapis.com/auth/spreadsheets.readonly",
    "https://www.googleapis.com/auth/presentations.readonly",

    # Gmail scope
    "https://www.googleapis.com/auth/gmail.send",
    "https://www.googleapis.com/auth/gmail.compose",
    "https://www.googleapis.com/auth/gmail.modify",

    # Tasks scope
    "https://www.googleapis.com/auth/tasks",
    "https://www.googleapis.com/auth/tasks.readonly",

    # Calendar scope (includes Meet creation)
    "https://www.googleapis.com/auth/calendar",
    "https://www.googleapis.com/auth/calendar.events",
    "https://www.googleapis.com/auth/calendar.events.owned",  # For creating events
]

def authenticate():
    """Authenticate and generate token.json with all required scopes."""
```

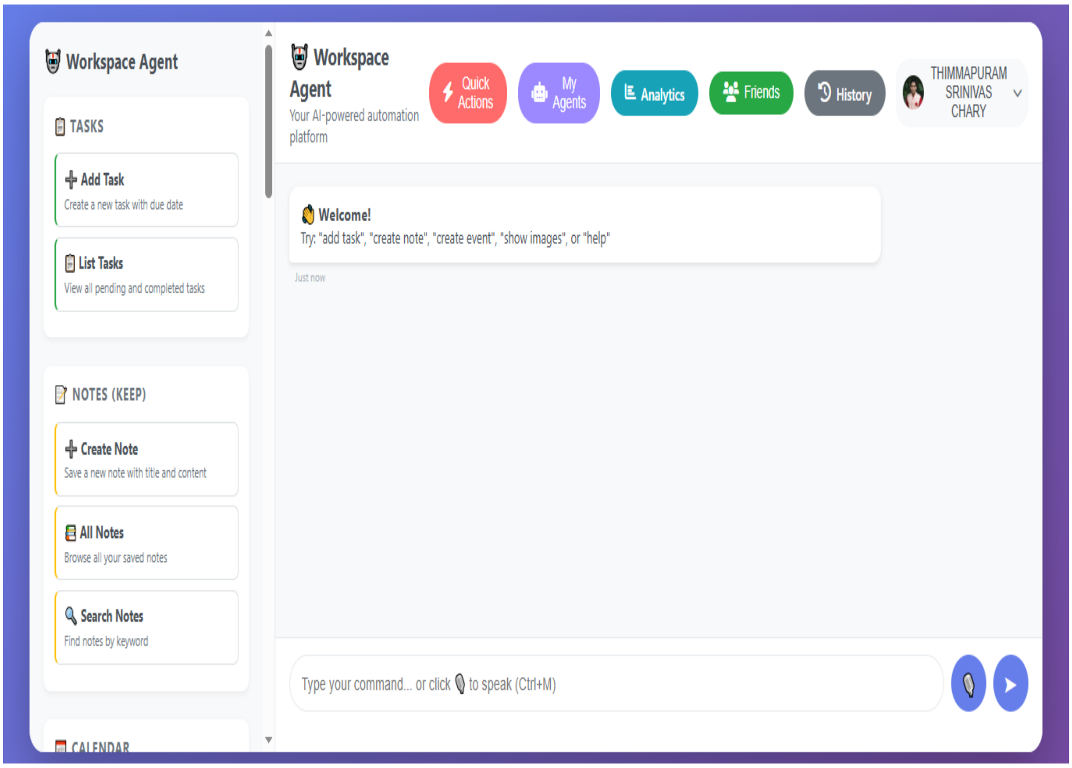
```
    creds = None
    token_path = "token.json"
    creds_path = "credentials.json"  # your OAuth client credentials file

    print("\n□ Workspace Agent Authentication")
    print("=" * 50)
    print("This will request access to:")
    print("  □ Google Drive - Read files")
    print("  □ Gmail - Send emails")
    print("  □ Google Tasks - Manage tasks")
    print("  □ Google Calendar - Manage events and create Google Meet")
    print("  □ Google Keep - Notes (local storage)")
    print("=" * 50)

    # Check if credentials.json exists
    if not os.path.exists(creds_path):
        print(f"\n❌Error: {creds_path} not found!")
        print("\nPlease download your OAuth client credentials from Google Cloud Console:")
        print("1. Go to https://console.cloud.google.com/")
        print("2. Select your project")
        print("3. Go to 'APIs & Services' > 'Credentials'")
        print("4. Create OAuth 2.0 Client ID (Desktop application)")
        print("5. Download JSON and save as 'credentials.json'")
        return
```

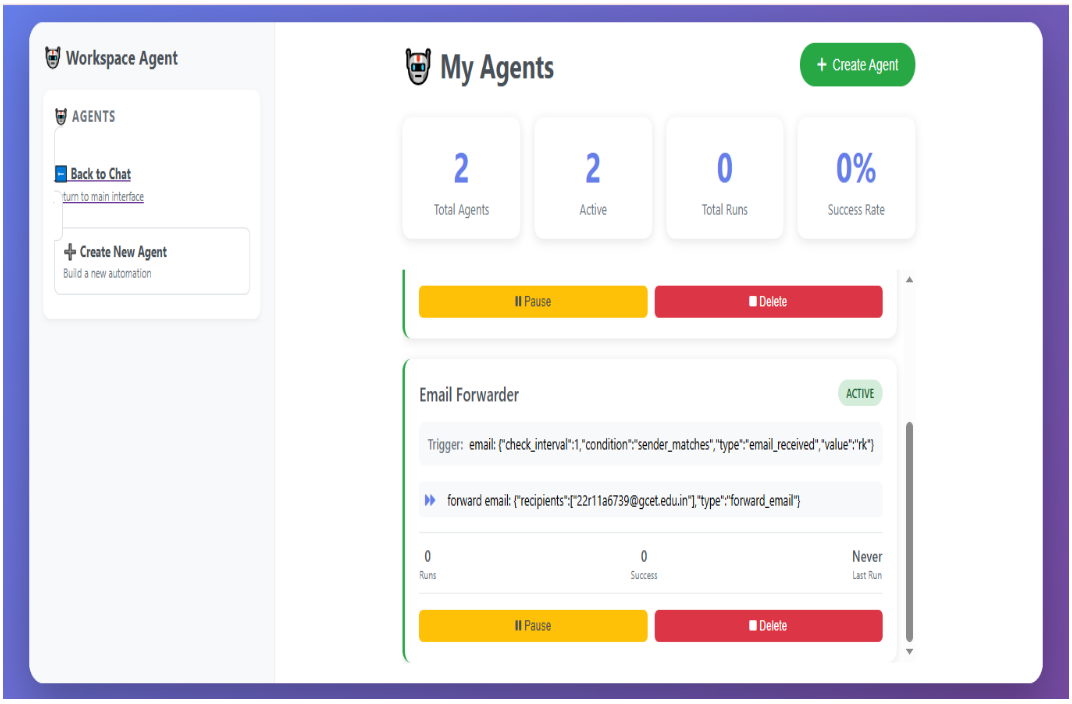
6. RESULTS

6.1 Home Page



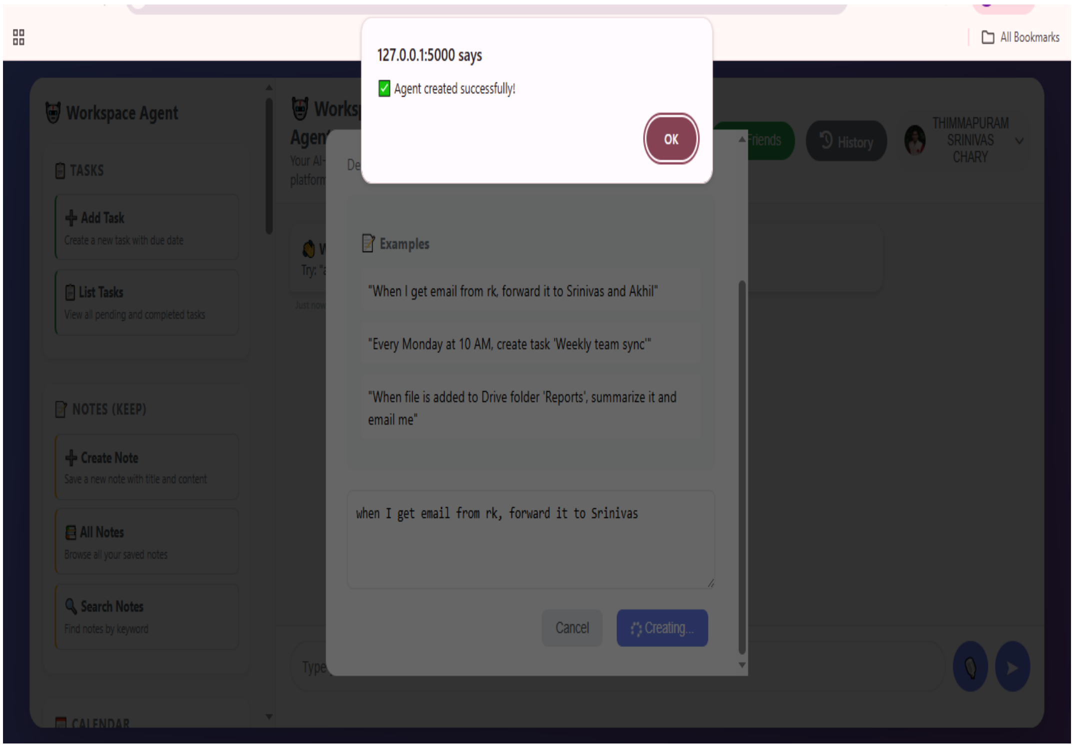
Screenshot 6.1.1 Home Page

6.2 My Agents Page



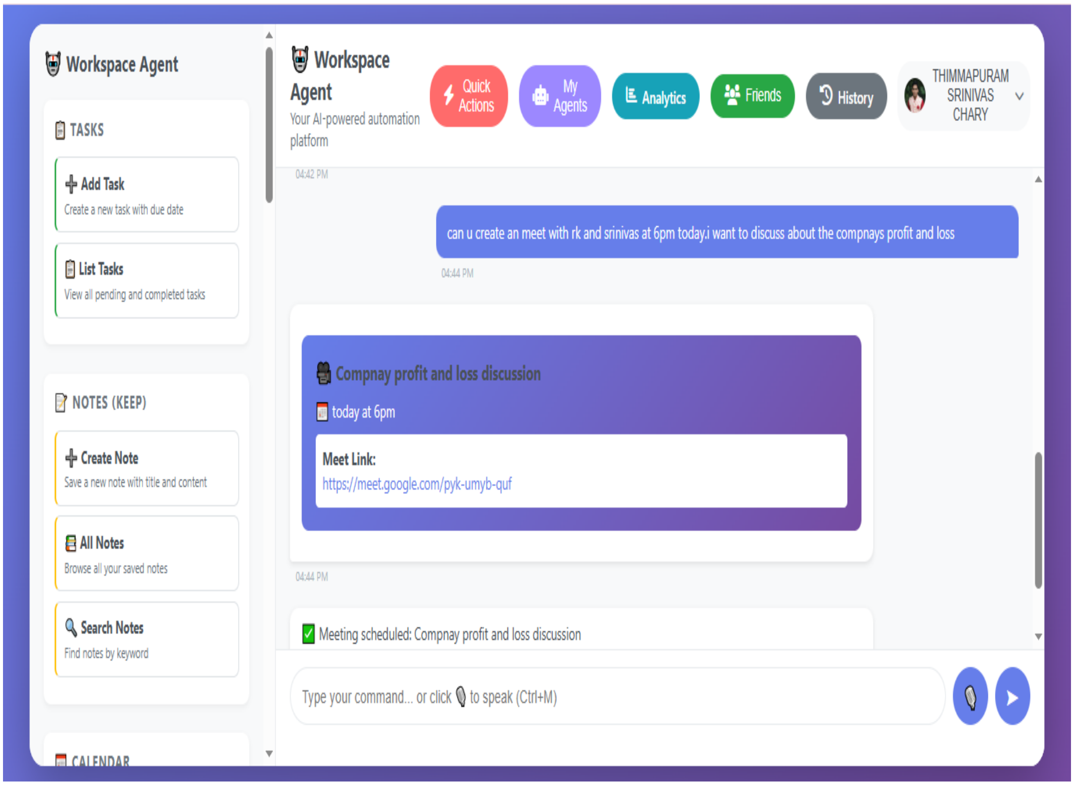
Screenshot 6.2.1 My Agents Page

6.3 Agent Creation Page



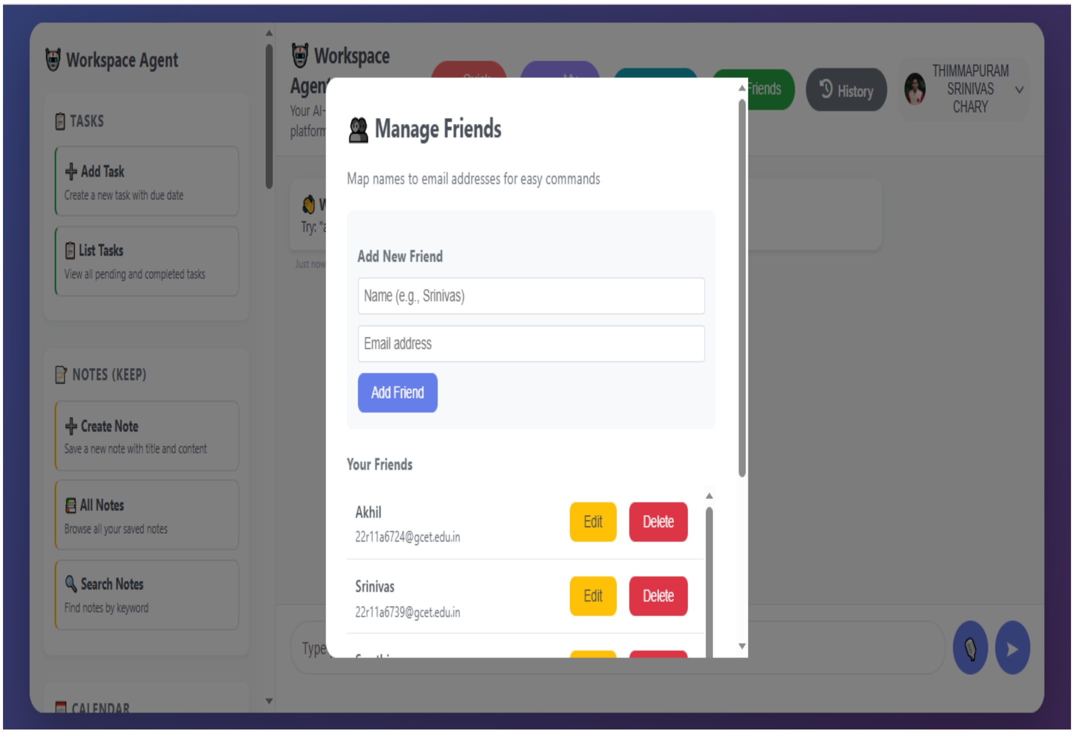
Screenshot 6.3.1 Agent Creation Page

6.4 Meeting Scheduler Page



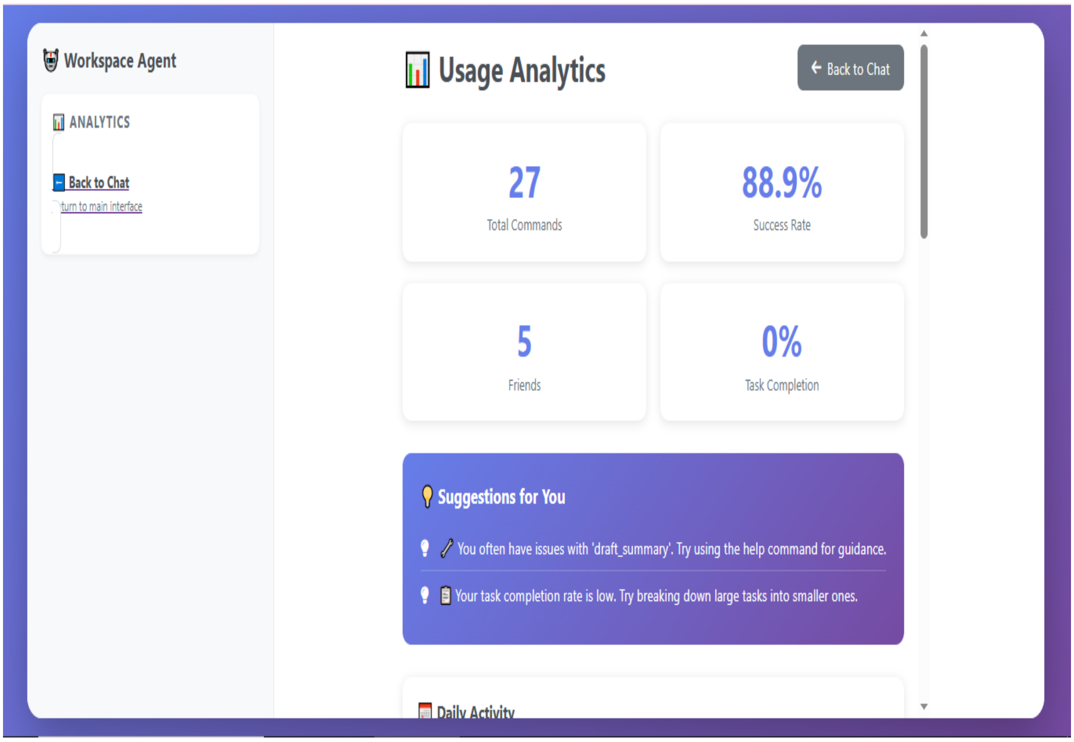
Screenshot 6.4.1 Meeting Scheduler Page

6.5 Manage Friends Page



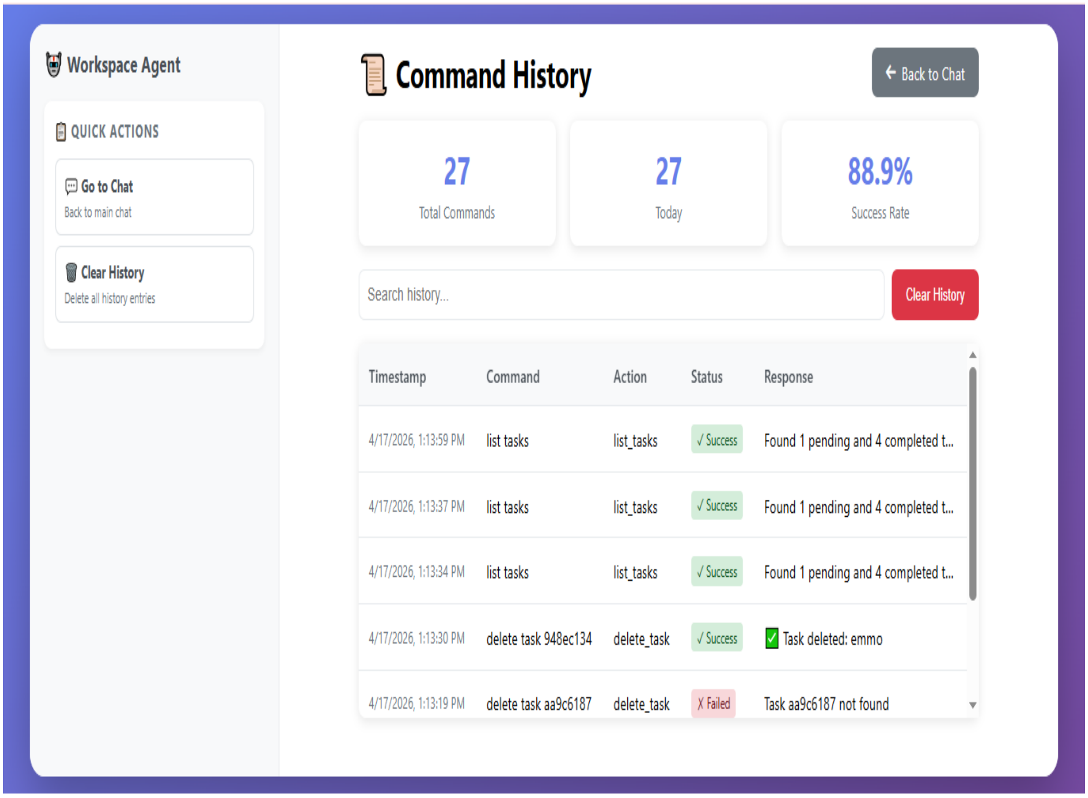
Screenshot 6.5.1 Manage Friends Page

6.6 Usage Analytics Dashboard Page



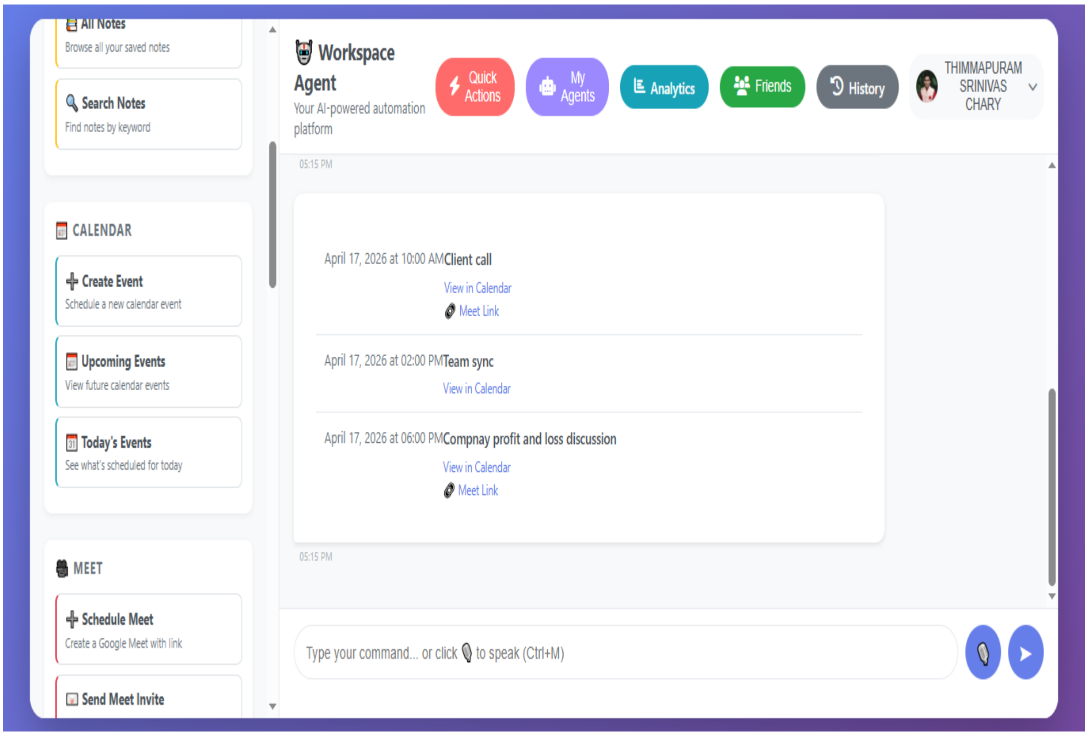
Screenshot 6.6.1 Usage Analytics Dashboard Page

6.7 Command History Page



Screenshot 6.7.1 Command History Page

6.8 Events List Page



Screenshot 6.8.1 Events List Page

7. TESTING AND TEST CASES

An essential component of creating the Unified AI Agent to Automate Tasks across Google Workspace is testing. It must also do things correctly in response to user instructions and integrate with other Google applications such as Gmail, Google Drive, Google Docs, Google Sheets and Google Calendar. The Unified AI Agent to automate tasks across Google Workspace is tested using types of inputs. They consist of a text command, application programming interface responses to requests to automate tasks and user interactions.

The following test cases are made to see if the Unified AI Agent for Automating Tasks, across Google Workspace is correct, efficient and reliable.

7.1 Command Processing Testing (NLP Model)

Test Case No	Name of the Test Case	Description	Expected Output	Actual Output	Pass/Fail
TC1.1	**Email Command**	**Input: “Send email to John”**	**Recognize email task**	**Task identified correctly**	**Pass**
TC1.2	**File Creation Command**	**Input: “Create a document”**	**Identify document creation**	**Document task detected**	**Pass**
TC1.3	**Schedule Command**	**Input: “Schedule meeting at 5 PM”**	**Identify calendar task**	**Calendar task detected**	**Pass**
TC1.4	**General Query**	**Input: “What is my schedule today?”**	**Retrieve schedule info**	**Info displayed correctly**	**Pass**
TC1.5	**Ambiguous Command**	**Input unclear command**	**Request clarification**	**Clarification asked**	**Pass**

Table 7.1 Command Processing Testing

7.2 Gmail Automation Testing

Test Case No	Name of the Test Case	Description	Expected Output	Actual Output	Pass/Fail
TC2.1	Send Email	Send email via command	Email sent successfully	Email sent	Pass
TC2.2	Read Emails	Fetch inbox messages	Display emails	Emails displayed	Pass
TC2.3	Search Email	Search by keyword	Show relevant emails	Correct results shown	Pass
TC2.4	Draft Email	Create email draft	Draft saved	Draft created	Pass
TC2.5	Invalid Email	Wrong email format	Error message	Error handled	Pass

Table 7.2 Gmail Automation Testing

7.3 Google Drive & Docs Testing

Test Case No	Name of the Test Case	Description	Expected Output	Actual Output	Pass/Fail
TC3.1	Create Document	Create Google Doc	Document created	Success	Pass
TC3.2	Upload File	Upload file to Drive	File uploaded	Uploaded successfully	Pass
TC3.3	Search File	Search file in Drive	File found	Correct file shown	Pass
TC3.4	Edit Document	Modify document	Changes saved	Updated successfully	Pass
TC3.5	Delete File	Delete file	File removed	Deleted successfully	Pass

Table 7.3 Google Drive & Docs Testing

8. CONCLUSIONS AND FUTURE WORK

8.1 Conclusion

The Unified AI Agent to automate tasks in Google Workspace is a considerable step toward combining Artificial Intelligence and productivity and collaboration tools. In a modern, busy, digital world, the management of various apps like Google Docs, Gmail, Google Drive, Google Calendar and Google Sheets can be inefficient, redundant, and low-productivity. The proposed system will solve these issues by proposing a centralized AI-based solution that would be able to automate routine operations, simplify the workflow process and improve the efficiency of the users. The system uses modern technologies like Artificial Intelligence, Machine Learning, Natural Language Processing, and Large Language Models to read user instructions, comprehend a scenario and perform actions in a smart way across different services in Google Workspace. The platform allows users to communicate with the system in a natural language, which means that it does not require them to perform complicated manual tasks and does not make it as reliant on technical skills. Additionally, the system provides real-time responses and intelligent suggestions, further enhancing decision-making and productivity. The project also shows how AI can be practically used to address real-world issues that have to do with task management and digital productivity. It throws light on the way smart automation can turn conventional processes into smart processes. Moreover, the modular and scalable nature of the system opens up opportunities to upgrade in the future, including connecting with other platforms, enhancing the contextual knowledge and predictive power. To sum up, the Unified AI Agent is an effective application to enhance productivity, manage tasks more easily, and ensure the effective use of Google Workspace services. It does not only simplify the amount of manual work, but also opens the path to the creation of smarter and more self-sufficient digital assistants, which form part of the intelligent working environment development.

8.2 Future Work

The Unified AI Agent to Automate Tasks in Google Workspace is a thing that can help us do our daily tasks on the computer. It is good at doing what we tell it to do.. There are a lot of things that we can do to make the Unified AI Agent better. We can make it smarter and more useful on platforms and for other people who use it. One thing we can do is help the Unified AI Agent understand what we are doing and what we like. Now it is very good at doing what we tell it to do.. In the future we can make it even smarter so it can see how we work and what we like to do. Then the Unified AI Agent can give us suggestions that're just for us. It can do tasks on its own without us telling it what to do. Another big thing we can do is make the Unified AI Agent work with tools like Microsoft Office, Slack and Zoom. This will make it more useful in life especially at work. We can also add some things to the Unified AI Agent like helping people work together at the same time. The Unified AI Agent can help many people work on the document, schedule meetings and make sure the team is working well. It is also very important to make sure the Unified AI Agent is safe and private. We can do this by making sure only the right people can use it and by keeping our information safe. We can use AI models to make the Unified AI Agent even better. This will make it more accurate, faster and better at making decisions. All these things will make the Unified AI Agent a powerful tool that can help us a lot. The Unified AI Agent will be very good, at helping us and making our work easier.

9. REFERENCES

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10. APPENDICES

The appendices section has further supportive information that supplements the development and implementation of the Unified AI agent to automate tasks across Google Workspace. This part comprises additional information about the system which adds to the knowledge about the system but they are not part of the main text of the report to avoid confusion and organization. There are several tools, technologies, and configurations used in the project and are recorded herein. These involve API configuration of the Google Workspace services like Gmail, Google Drive, Google Docs, Google Sheets and Google Calendar. The steps of authentication based on OAuth 2.0, key generation, and permission control are detailed to provide a secure and seamless integration of the AI agent and the Google services. Further, the input commands to the sample and the system output is given to illustrate how people can communicate with the AI agent in a natural language. These are just a few instances of how the system can understand user intent and execute tasks like emailing, schedules, documents creation, and file organization automatically. Screenshots of user interface and system responses are also provided to provide a clear visual depiction of the functioning model. The appendices also include data regarding the datasets, testing conditions and performance evaluation measures that were employed in the development stage. This involves test cases that are used to confirm the accuracy, response time and reliability of the system in various tasks and conditions. Any restrictions noted in testing are also briefly recorded. Besides, there are code snippets and details of system configuration that are provided to give technical understanding on how the AI agent was implemented. This information can be helpful in any way to make improvements or repeat the work or conduct a new research. On the whole, the section serves as the overall reference that contributes to the project providing further clarity, transparency, and technical depth without excessive length and focus of the main report.